StabiliRead: A Gaze–Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Middle-Aged and Older Adults

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Abstract

Digital reading is now one of the main ways people access information. Unlike traditional reading, digital environments are dynamic. Text is often surrounded by other on-screen elements, such as notifications and pop-ups, which can interrupt how we read. Readers, particularly middle-aged and older adults, often experience difficulty navigating this environment due to their increased susceptibility to distraction and slower processing speeds, resulting in increased cognitive effort and reduced reading efficiency. This study aims to address the lack of adaptive graded-based systems that support recovery during disrupted reading. To address this gap, the study proposes a personalized gaze-driven adaptive reading interface that provides graded visual scaffolding based on moment-to-moment reading behavior. The system is implemented as an extension of NewsMead, an existing Android news-reading application designed for older Filipino adults. This system introduces a stability index derived from gaze metrics to detect instability and dynamically adjust visual support in order to assist re-orientation and maintain reading continuity. The study will be conducted using eye-tracking to capture gaze behavior and identify instability indicators, and evaluate system effectiveness through measures such as fixation duration, saccades, and regression, indicating stability and recovery latency. By focusing on middle-aged and older adults (aged 45 and above), where reading instability is often more pronounced the study aims to provide a clearer evaluation of recovery-focused support and contribute to a more responsive and human-centric digital reading systems.

Keywords: digital reading, gaze-based interaction, visual scaffolding, reading stability, adaptive interfaces

Contents

List of Figures

List of Tables

Chapter 1

Introduction

With the increasing reliance on digital technologies for reading, particularly among middle-aged and older adults, understanding how aging affects reading performance has become increasingly important. Reading in general is not just a visual activity, but rather a complex, multifaceted cognitive process that requires continuous distribution of attention, language processing, and thought. That said, reading is a cognitive process governed by lexical access, which enables the recognition and retrieval of word meanings (White, Palmer, Boynton, & Yeatman, 2019). In retrospect, lexical access unfolds through eye movements, as directly reflected in gaze-behavioral patterns such as prolonged reading, saccadic patterns, and frequent regressions (White et al., 2019; Pagán et al., 2025). As such, age accentuates these patterns, often leading to slower processing speed or longer fixation durations, particularly during reading disruptions (Pagán et al., 2025; Li et al., 2019; Liu, Patel, & Kwon, 2017; Paterson, McGowan, & Jordan, 2013). However, most existing studies focus heavily on outcome-based metrics such as overall comprehension and reading speed but do not examine moment-to-moment reading sessions, thereby overlooking the recovery phase, during which readers reorient themselves after reading disruptions. In line with this, the research highlights a critical gap for middle-aged and older adults: **age-related changes in cognitive and visual processing amplify the recovery process required during prolonged reading sessions**.

1.1 Background of the Study

The use of digital reading in contemporary society has become rampant. With the widespread use of smartphones, tablets, and computers, reading has transformed from primarily being print to digital. That said, research in eye-tracking and reading science has shown that reading depends on coordinated eye movements, including fixations and saccades, which reflect underlying cognitive processing (White et al., 2019; Pagán et al., 2025). Studies have consistently demonstrated that aging influences these oculomotor patterns, with older adults exhibiting longer fixation durations, more frequent regressions, and slower processing speeds. Although comprehension accuracy may remain relatively stable, recovery from disruptions requires greater effort, especially in the older population. With the massive shift of literacy to digital platforms, it has expanded to various age groups, particularly older adults, where they become more reliant on accessing information. Furthermore, as the digital media continue to replace traditional print, aging individuals find themselves struggling with a combination of factors such as age-related physical changes, cognitive changes, and particularly a poor design interface that fails to accommodate their cognitive and accessibility needs (Pagán et al., 2025; Li et al., 2019; Liu et al., 2017).

Furthermore, existing reading digital tools have started to introduce various forms of visual guidance to support reading performance. For example, BeeLine Reader, an assistive technology tool that uses color gradients to guide eye movement during on-screen reading, allows users to read faster and more easily. These types of digital reading tools show early attempts at visual guidance systems and show a promising result of visual scaffolding in terms of reading support. However, these systems remain largely static and do not adapt dynamically to the reader's moment-to-moment cognitive state or reading instability.

The problem in today's current digital reading environments is that they largely assume a uniform reading behavior, which fails to account for how aging alters the dynamic process of reading, specifically under conditions of instability. Beyond reading comprehension, one significant factor that also needs to be addressed is the recovery latency when it comes to reading. When lexical processing is interrupted, readers must reorient and re-establish stable forward progression. For older adults, age-related changes in visual perception and processing speed may increase the time and effort required to regain reading continuity, potentially amplifying cognitive fatigue during sustained digital reading sessions (Pagán et al., 2025; Li et al., 2019). With that being said, with the regressive aspect that comes with age, regaining that processing sequence can further amplify cognitive effort and fatigue for the aging population.

This problem highlights a critical shift in perspective, wherein, rather than basing reading performance on comprehension outcomes of the reader, the shift focuses on examining moment-to-moment stability of reading and the processes involved in recovering from reading disruptions. In hindsight, the research domain is directed to the aging population and how their digital reading experience can be enhanced. Through real-time eye tracking, the study is situated in developing a gaze-driven digital adaptive reading interface for middle-aged and older adults in order to provide a more stable reading progression and reduce the overall cognitive load that comes with age. As more adults increasingly rely on digital platforms for reading and information access, there is a growing interest in developing an interface that can accommodate age-related influences such as processing speed, attention stability, and visual perception (Li et al., 2019; Liu et al., 2017)

Through real-time eye tracking, we can measure the gaze-behavior of the reader, such as fixation patterns, regressions, and saccadic eye movements, which are highly indicative of the lexical process when it comes to reading (Pagán et al., 2025; Shojaeizadeh & Djamasbi, 2018). As such, rather than focusing on the comprehension outcomes of the reader, the study emphasizes finding sustainable ways to support reading stability and reduce the amount of re-orientation during sustained digital reading of middle-aged and older adults. However, it is worth noting that this study points to an interesting question: **how can reading interfaces move beyond passive observation of gaze behavior and, alternatively, respond to detected disruptions in real-time that are specifically tailored to each reader?**

A considerable number of studies have demonstrated reliable indicators of lexical access, with fixation durations, saccading eye movements, and regression frequency being the key measures of the required effort of semantic integration (Pagán et al., 2025; Payne et al., 2020; Li et al., 2019; White, Palmer, & Boynton, 2020). In light of this, more research has shown that older adults have been found to exhibit longer fixation durations, slower processing speed, and frequent regressions, yet the ability to comprehend remains intact, but efficiency and effectiveness are compromised. In retrospect, the relationship between adaptive reading interfaces and gaze-contingent systems is not new and has been widely explored, in which many systems dynamically modify text presentation in real-time. These systems aim to improve the reading experience by improving engagement, reducing mind-wandering, and overall enhancing reading comprehension across various population groups (Li et al., 2019).

However, existing studies heavily rely on outcome-based measures of readers such as reading speed and comprehension accuracy, which overlook the underlying recovery processes that occur during reading sessions. In addition, existing systems often rely on fixed thresholds or generalized rules to trigger adaptations, which limits individual variability and the continuous effect of reading instability in reading sessions. With that being said, what this study aims to do differently is that, rather than focusing on outcome-based metrics of the reader, it focuses on **a baseline-calibrated**, stability-driven adaptive reading interface that reduces recovery latency and supports stable forward progression during sustained digital reading in middle-aged and older adults.

Much research using eye tracking technology has shed light on the nuances between oculomotor characteristics and cognitive capabilities in reading situations. Comparatively less research has been conducted on the recovery phase following reading instability and disruptions. During sustained digital reading, disruptions such as regressions or attentional drifts require readers to reorient themselves in order to establish stable reading once more. This recovery process can impose additional cognitive strain, particularly among older adults who exhibit cognitive and oculomotor constraints leading to reading behaviors such as longer fixation durations and frequent regressions.

Although current gaze-contingent reading interfaces aim to improve engagement and performance, most systems evaluate overall reading outcomes such as speed or accuracy rather than directly measuring recovery latency or modeling instability as a graded phenomenon. Moreover, current scaffolding techniques are typically static or triggered by fixed thresholds, lacking baseline-calibrated, proportional adaptation to moment-to-moment instability.

Taking all these together, the limitations point to a clear research direction where there is insufficient focus on modeling and giving emphasis on the recovery phase of reading, which also serves as a dynamic and important process of achieving lexical access within adaptive digital reading systems. Therefore, this study aims to evaluate **whether a baseline-calibrated, adaptive gaze-contingent visual scaffolding interface can reduce recovery latency and enhance reading stability during sustained digital reading among middle-aged and older adults.**

To represent moment-to-moment changes with regards to the reading stability, this study proposes an Adaptive Instability Index (AII) which is a continuous, participant-relative measure derived from gaze behavior. The AII combines recent changes in regression rate, dwell-time, and fixation rate relative to each reader's established baseline. Higher AII values indicate stronger evidence of reading instability, whereas lower values indicate behavior closer to the reader's typical reading pattern. The adaptive interface uses the AII to determine the appropriate level of graded visual scaffolding. The detailed computation, component weighting, temporal smoothing, and scaffold-level thresholds of the AII are presented in Chapter 4.

1.2 Objectives of the Study

The general objective of this study is to design a baseline-calibrated, gaze-driven adaptive reading interface aimed at reducing recovery latency during sustained digital reading among middle-aged and older adults. By leveraging real-time eye-tracking data, the system seeks to detect reading instability based on measurable gaze behaviors and provide graded visual scaffolding to support efficient re-orientation and stable forward progression.

To achieve this general objective, the study is guided by the following specific objectives:

1. To identify gaze-based indicators of reading instability during sustained digital reading, including fixation duration, regression frequency, saccade patterns, and reading pauses;
2. To define recovery latency as a measurable outcome using gaze-based metrics such as the time required to re-establish stable reading following a disruption;
3. To develop a baseline-calibrated adaptive reading interface that provides graded visual scaffolding based on deviations from user-specific reading behavior;
4. To implement a trigger-based adaptation mechanism, where interface changes occur only when instability exceeds predefined thresholds within a sliding time window;
5. To evaluate the effect of the adaptive interface on recovery latency compared to a static digital reading interface;

1.3 Scope and Limitations of the Research

1.3.1 Scope of the Study

This study focuses on the design and evaluation of a gaze-driven adaptive reading interface aimed at reducing recovery latency and improving reading stability among middle-aged and older adults during long-form digital reading. The research centers on the use of eye-tracking data as a behavioral indicator of reading dynamics, particularly in identifying moments of instability and subsequent recovery. Gaze features such as fixation duration, regression patterns, and dwell times are used to model reading stability and to operationalize recovery latency as a measurable construct.

The proposed system is implemented as an extension of NewsMead, an existing Android news-reading application for older Filipino adults called NewsMead that provides a participatory-design-validated reading interface (Tiongquico et al., 2024). The primary deployment platform is an Android smartphone with a front-facing camera, which serves as the gaze-input device. This platform was selected for its accessibility, its alignment with how the target population typically accesses digital news, and the availability of an existing, validated reading interface on which the adaptive layer is built. A possible extension to other Android devices is planned for future work as the system matures.

For evaluation, The reading materials used in the study consist of news articles drawn from the NewsMead platform, reflecting the real-world digital reading context of the target population. A standardized font size will be applied across all participants to prevent typography from confounding gaze-metric comparisons.

In the Philippines and under Republic Act No. 9994 (Congress of the Philippines, 2010), *older adult* and *senior citizen* refer to persons aged 60 and above. This study deliberately spans a wider band, beginning at 45, because the visual and oculomotor changes it targets, presbyopia and the decline in contrast sensitivity, begin around the fifth decade rather than at 60. Throughout this document, *middle-aged and older adults* refers to the study population of adults aged 45 and above, whereas *older adults* is reserved for reporting prior work whose samples were drawn from the 60-and-above group. The distinction matters because much of the reviewed literature studied that narrower group, and its findings are not assumed to transfer unchanged to readers in their forties and fifties. NewsMead itself was designed for older Filipino adults in the narrower sense; the adaptive layer evaluated here is assessed across the wider band, on a reading surface that is unchanged for all participants.

Our participants will be adults aged 45 and above, with no upper age limit imposed. They will be screened based on their educational attainment and familiarity with digital devices, requiring them to have completed at least secondary education. Vision will be confirmed normal to corrected-to-normal through a near visual acuity test, and participants with diagnosed ocular pathology affecting reading, severe visual impairments, or neurological disorders that would substantially affect reading behavior will be excluded. Suitability at the upper end of the age range is therefore established by screening on vision and reading capability directly, rather than by an age cutoff. This is to ensure that observed reading behavior reflects the natural variation within the target age group rather than clinically atypical conditions.

System performance is assessed through a mixed-methods evaluation combining gaze-based behavioral metrics, primarily recovery latency, with a qualitative post-session assessment of participants' experience with the adaptive interface.

1.3.2 Limitations of the Study

This study is subject to several limitations. The research focuses specifically on adult readers aged 45 and above, and findings may not generalize to younger populations or individuals with clinically diagnosed cognitive or reading impairments.

The system relies on gaze data obtained through a consumer-grade Android front-facing camera, which introduces variability and reduced spatial precision compared to specialized research-grade eye-tracking hardware. In addition, gaze based models are very limited for android devices.

This precision limit constrains the spatial granularity of the adaptive scaffolding. Measured vertical error on the target device is comparable to the height of a single rendered line of text, and horizontal error is larger still, so the system is designed to identify the region of text a participant is reading rather than an individual word. Scaffolding is defined at the level of a word region, a line, or a larger focus area accordingly, and finer-grained intervention is outside the scope of this work. Reliable line-level and region-level attribution is treated as a claim to be verified during pilot testing rather than as an established capability, and the accuracy figures on which these bounds are based were obtained from members of the development team rather than from participants in the target age range.

Recovery latency is defined using gaze-based behavioral measures, which serve as indirect proxies for cognitive processes and may not fully capture deeper aspects of comprehension and mental effort.

The reading materials are drawn from a single platform (NewsMead) (Tiongquico et al., 2024) and consist of news articles in a specific language and cultural context. These materials may not fully represent the diversity of real-world digital reading contexts, such as casual browsing, multimedia content, or highly domain-specific texts.

Furthermore, the evaluation is conducted in a controlled experimental setting, which may not fully reflect real-world environments where distractions, device variability, and user habits influence reading behavior and gaze tracking performance.

Cognitive load is inferred from gaze-derived indicators and qualitative self-report only, without the use of direct physiological measurements such as EEG or heart rates monitoring.

Finally, gaze-tracking performance for participants who wear reading glasses or have presbyopia may differ from younger participants, as lens properties and eye-muscle characteristics affect iris landmark detection and calibration accuracy. Empirical validation of the tracker's performance on the specific target population and hardware is therefore a prerequisite before full data collection.

1.4 Significance of the Research

This study contributes to the field of Human-Computer Interaction by shifting the focus of gaze-based reading systems from attention detection and performance measurement toward supporting recovery and maintaining reading continuity. While existing systems primarily rely on static, non-adaptive interfaces and discrete detection of attention lapses, this research introduces an adaptive approach that responds dynamically to the reader's state.

The study proposes a gaze-driven interface that models reading behavior continuously and translates it into graded levels of visual scaffolding. This enables the system to provide context-sensitive support, ranging from minimal guidance during stable reading to stronger re-entry assistance during periods of instability. By doing so, the system moves beyond binary or static interventions and instead offers progressive, multi-level adaptation based on real-time gaze behavior. From an evaluation perspective, the study introduces recovery latency and reading stability as key metrics, rather than focusing solely on speed and comprehension. This provides a more nuanced understanding of how readers recover from attention lapses and how interface adaptations can support this process. Furthermore, this research places particular emphasis on middle-aged and older adults, a population that is often underrepresented in gaze-based interface design. By addressing age-related challenges in reading stability and recovery, the proposed system aims to improve accessibility and support more inclusive digital reading experiences.

Overall, this study highlights the potential of adaptive, graded visual scaffolding as a novel approach for recovery-focused reading support, contributing both a conceptual framework and a practical system design for real-time, gaze-driven interaction.

1.5 Research Methodology

This study applies a mixed-methods approach in order to investigate how a gaze-driven adaptive reading interface can support reading instability and recovery for middle-aged and older adults. The methodology combines a quantitative behavioral assessment with comparing adaptive vs static interfaces with a qualitative evaluation on user experience. This allows the study to be able to measure the system's effectiveness while understanding adult readers' experiences and preferences with regard to the adaptive scaffolding model.

The study is conducted in two phases. The first phase is the formative phase, in which the system's overall design is validated with a small group of participants before the main evaluation. The second phase is the evaluation itself in which the adaptive interface is formally compared against a static reading interface. Figure 1.1 illustrates the overall flow of the summative phase.

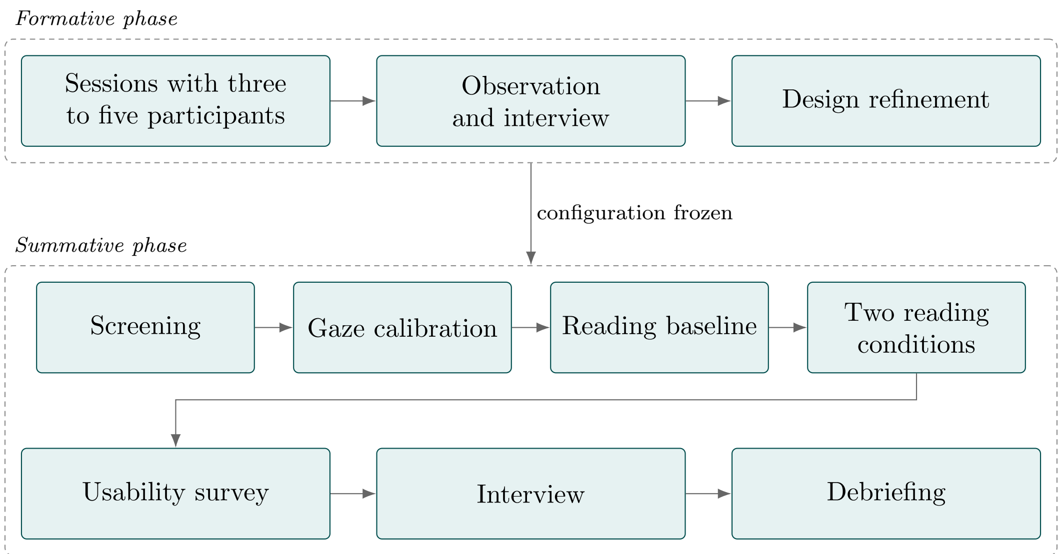

Figure 1.1: Research methodology flow of the proposed system. The two reading conditions, adaptive and static, are presented in counterbalanced order, with comprehension questions after each. The closing steps follow a fixed order, as the debriefing would otherwise influence the interview.

In the formative phase, three to five participants from the target population interact with an early version of the adaptive interface under the researcher's observation. Following each session, a brief interview is conducted to surface usability concerns, confusing interactions, and reactions to the scaffolding interventions. Findings from this phase inform design refinements before the main evaluation begins. This iterative validation step ensures that the system is appropriately calibrated to the needs and expectations of the target population prior to formal data collection.

In the summative evaluation phase, participants are profiled by age, reading level, vision conditions, and familiarity with digital devices. This is done through a short vision check to confirm normal or corrected-to-normal vision, a reading task to assess basic reading capability, and a background questionnaire to collect demographic and device usage information. The study includes individuals aged 45 and above with normal or corrected vision, while excluding those with severe visual impairments or neurological disorders that may significantly affect reading behavior. This is to ensure that observed reading behavior reflects the natural variation within the target age group rather than clinically atypical conditions.

Before the main reading tasks, a baseline calibration phase will be conducted to establish each participant's typical reading behavior. During this phase, the participants will read standardized text passages under normal conditions while gaze data is collected. Using the baseline measure an individual reading profile will be constructed, which serves as a reference for detecting deviations in users' reading stability. This approach allows the system to determine instability relative to each participant's normal behavior rather than relying on fixed thresholds.

Reading behavior during the experiment will be characterized using measurable indicators such as fixation duration, regression count, reading pauses, and gaze-dispersion. These indicators will be used to derive key performance metrics including reading speed in words per minute, comprehension scores based on follow-up questions, regression frequency, and fixation duration. In this study, recovery latency will be the central metric in the system defined as the time interval between a disruption event, such as a loss of reading position and the point at which stable reading patterns are re-established. The initial analysis will focus on fixation duration and regression rate as primary indicators of instability, while additional exploratory metrics such as recovery latency and stability variance will be used to further examine reading dynamics and find the appropriate parameters.

A continuous gaze-derived stability index will be computed in real time by aggregating gaze features within a sliding time window. Instability is identified when deviations from baseline exceed predefined thresholds, such as a significant increase in fixation duration or a spike in regression frequency. Based on this stability index, the system implements a trigger-based adaptation mechanism. As illustrated in Figure 1.2, the system maps different levels of instability to corresponding levels of visual support. The interface does not change continuously; instead, visual adjustments are applied only when instability is detected. For low levels of instability, minimal interventions such as word-region highlighting are applied, while higher levels trigger stronger support such as line emphasis, focus windows, and re-entry assistance. These graded scaffolding techniques are designed to help the reader maintain orientation and recover their position in the text.

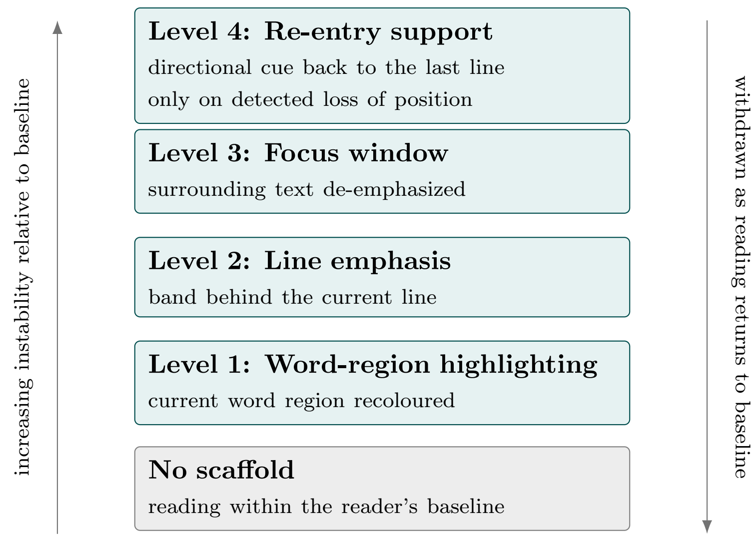

Figure 1.2: Graded visual scaffolding driven by detected reading instability. Support escalates one level at a time as instability persists and is withdrawn as reading returns to baseline, so that only one level is active at any moment. The strongest level is reached only when a loss of reading position is detected; high instability without that evidence remains at Level 3.

In order to evaluate the system's effectiveness, participants will perform reading tasks with the adaptive system and without it. To minimize learning effects, the order of these conditions will be controlled by assigning participants to different sequences. Performance across conditions will be compared using recovery latency, reading stability, comprehension, and cognitive load. This comparison allows the study to determine whether adaptive, stability-driven support improves reading continuity and reduces the effort required to recover from disruptions.

Throughout this study, established ethical guidelines will be followed for research involving participants. First and foremost, participation will be completely voluntary and informed consent will be obtained prior to data collection. The participants will be informed of the study procedures and their right to withdraw at any time without penalty. Moreover, no personally identifiable information will be collected, and all the recorded data shall be anonymized and used solely for research purposes. In order to ensure the protection and safety of the elderly, the reading task durations will be controlled in order to prevent fatigue.

Chapter 2

Related Works

2.1 Age-Related Reading Behavior and Oculomotor Characteristics

Reading is a collaborative effort of the eyes' oculomotor response and the mind's cognitive prowess; processing visual information through a sequence of fixations characterized as a brief pause of the eye on a specific word or location in text and saccades which are rapid eye movements between fixations. Skilled adult readers typically exhibit fixation durations averaging approximately 200-250 ms during initial readings (Pagán et al., 2025). Behavioral patterns exhibited by these individuals are 2.6 regressions per sentence likely attributed to text complexity or ambiguity as well as comprehension difficulty (Pagán et al., 2025). They also have a probability of skipping around 21-27% of the words mostly due to word length and contextual knowledge (Pagán et al., 2025). This is assumed as a way to initiate linguistic processing in advance, before encountering the word using previous contextual experiences (Veldre et al., 2020). In comparison, Older adults tend to exhibit longer fixation durations on average, around 263ms compared to the 232ms for younger adults and produce significantly more regressions at around 3.5 per sentence (Moreno et al., 2018). This implies that when older adults encounter a word that requires integration or reanalysis, they spend significantly more time re-reading and resolving it before continuing, compared to younger adults (Moreno et al., 2018).

The perceptual span is the window during which an individual extracts information at a single fixation. During each pause, you don't just see the word you're directly looking at. You are also gathering information from nearby words. In alphabetic languages, particularly English, skilled readers have been observed to perceive 3-4 letters to the left and about 14-15 letters to the right so the span is asymmetric (Veldre et al., 2020). This is likely attributed to the structure of the English language, which is read from left to right. The rightward span is typically larger than the left in most cases while lower readers may asymptote with smaller windows (White et al., 2020). An interesting note, older adults exhibit reduced asymmetry in perceptual span, characterized by greater leftward processing and a rightward span that depends on reading proficiency (Veldre et al., 2020). Specifically, they have been observed to have similar right spans with younger adults, but the left is marginally higher at about 9 letters in comparison to the 3-4, and for older adults with lower literacy rates, they perceive 9 letters from both left to right (Veldre et al., 2020). The expanded leftward perceptual span observed in older readers may serve as a compensatory mechanism to facilitate re-processing of previously read text, potentially reducing the cognitive load associated with recovery from processing difficulties (Veldre et al., 2020). This indicates that they have much reduced asymmetry and that perceptual span is not fixed and varies across different proficiency and age groups. The perceptual span is important to our research because it determines how much text a reader can effectively process at once, which allows our adaptive gaze-based interface to personalize visual support based on each user's real-time reading capacity.

Now, when it comes to the speed of processing the meaning behind the text, skilled readers exhibit systematic patterns in relation to the lexical properties of words. High-frequency words are processed far quicker than less common words. This is observed through short fixation durations and a higher probability of being skipped entirely (Li et al., 2019). Moreover, words that are highly predictable from prior sentence context are also processed more efficiently (Whitford & D., 2017). Older adults in comparison exhibit longer durations and more frequent fixations per unit of text, indicating reduced information extraction per fixation (Moreno et al., 2018). Older adults experience greater difficulty processing rare words, indicating slower access to word meanings during reading (Li et al., 2019; Liu et al., 2017). Text presentation has also been noted as a primary factor in lexical processing for older adults, where they experience greater disruptions from more condensed text spacing, with extra-condensed text (20% smaller inter-letter and inter-word spacing) causing significantly longer reading times and regressions (Li et al., 2019). This heightened sensitivity to visual crowding indicates age-related declines in visual processing efficiency (Li et al., 2019; Liu et al., 2017).

It should be noted, however, that this decline in visual processing efficiency is not just a matter of slower eye movements. It also comes from a specific, well-documented change in the aging eye itself. Presbyopia, being one of these changes, begins to appear symptomatically between ages 40 and 45 and by 60, becomes nearly universal. In effect, this change or condition progressively reduces the eye's ability to focus on near objects as the lens loses flexibility due to age-related protein cross-linking (Singh, Zeppieri, & Tripathy, 2024). Alongside this, contrast sensitivity also declines with age. Sensitivity to high spatial-frequency detail, the kind needed to resolve fine print and other smaller details, starts to decline between ages 40 and 50. By ages 50 to 60, this decline spreads to all spatial frequencies, not just the high ones (Owsley, Sekuler, & Siemsen, 1983). These studies show that this decline happens gradually rather than suddenly appearing later in life. A large study using the MNREAD reading vision test found a steady, linear decline in maximum reading speed and reading accessibility from 20 to 80. This study found that the decline worsens under dim lighting (Peñaloza, Goddin, Friedman, Owsley, & Kwon, 2025)

Moreover, these visual changes do not just show up in vision tests; they also show up in how people actually read. One study examined presbyopic adults with a mean age of 48.9 years. When the researchers added visual stress with monovision correction, fixation counts and dwell times increased significantly, but only during the reading of effortful, unrelated word and nonword arrays. No such change occurred during natural reading of full sentences (Zeri, Naroo, Zoccolotti, & De Luca, 2018). This suggests dwell-related metrics respond specifically to presbyopic strain, but mainly under demanding visual conditions. Another study directly compared young (18–22), middle-aged (41–51), and older (65–79) readers. Under normal, undegraded text, middle-aged adults' fixation durations (237ms) and regression rates (2.3 per sentence) closely matched those of young adults (236ms and 2.4 per sentence). But disruption from reduced-contrast text increased with age across the sample (Warrington, McGowan, Paterson, & White, 2018). This indicates that readers in their forties and fifties read much like younger adults under good viewing conditions, but fall behind specifically when viewing conditions degrade, the same kind of degradation presbyopia and contrast-sensitivity decline would produce on a screen.

Further studies have also explored the cognitive and oculomotor costs associated with error recovery and reanalysis processes, focusing on the duration of the regression path and evidence of greater recovery efforts and recovery behavior (Moreno et al., 2018). Empirical studies provide clear evidence that older readers require more time and effort to resolve processing difficulties and re-establish comprehension (Moreno et al., 2018). This is guided by a number of observations. Regression path duration (RPD), defined as the sum of fixations on a region until the eyes move past it to the right, captures the time spent re-reading and re-integrating information before moving forward (Moreno et al., 2018). In this study, older adults demonstrate significantly more regressions with syntactically complex sentences, having a RPD of hundreds of milliseconds longer than younger adults (Moreno et al., 2018). This shows that when ambiguity was introduced older adults are seen to require longer periods of recovery time. Recovery is not only triggered by syntactic complexity but also by difficulties in integrating words that are unexpected within a sentence context (Pagán et al., 2025). Both young and older adults benefit from predictable words. However, older adults show much larger slowdowns when a word is less predictable (Pagán et al., 2025). This suggests that older adults rely more heavily on contextual prediction and when prediction fails, recovery requires a greater amount of effort in comparison (Pagán et al., 2025). While recovery dynamics are well documented in controlled sentence-level tasks, there remains limited research quantifying re-orientation costs in long-form digital reading contexts, representing a critical gap in the literature.

While existing literature provides extensive evidence on age-related differences in fixation duration, regression behavior, perceptual span, and recovery dynamics, most findings are derived from controlled, sentence-level experiments. These studies primarily quantify processing difficulty and recovery effort but do not translate these insights into adaptive interface design. In particular, there is limited work that operationalizes recovery cost in long-form digital reading or leverages these behavioral markers to guide real-time adaptive support. This gap motivates the need for systems that not only detect reading difficulty but also actively support recovery and re-orientation in sustained reading contexts, particularly for older adults.

2.2 Gaze-Contingent and Adaptive Reading Interfaces

While section 2.1 covers the dynamics of age-related reading behaviors, establishing that age groups have different oculomotor and cognitive capacities, these changes do not occur in isolation from the reading environment. Digital display parameters such as spacing, contrast, text-size, and scrolling interact directly with reading performance. Studies have shown that increases in age result in higher visual crowding sensitivity. The most pronounced finding is that with condensed text, particularly at -20%, readers experience the longest reading times and fixation durations, as well as the highest regression frequency, with older adults experiencing 20-30% more disruptions compared to younger age groups (Li et al., 2019; Liu et al., 2017). This indicates that display density directly amplifies reading instability and recovery burden may increase under condensed text spacing. In a similar fashion, display settings such as luminance and contrast, particularly in low levels, have shown to result in 14-22% slower reading in relation to younger adults(Paterson et al., 2013). Researchers have also identified exactly why lower contrast slows reading. When they reduced text contrast alone, with no aging or cognitive factors involved, fixation duration increased by 68 percent, and regressive saccades increased by 19 percent. Together with saccade amplitude, these measures explained 90 percent of the change in reading speed (Yu, Shamsi, & Kwon, 2022). This shows that lower contrast directly changes the same gaze measures this study uses to detect reading difficulty, even when nothing is cognitively wrong. Scrolling interfaces common in digital form reading, introduce oculomotor and cognitive demands that result in different behavioral patterns across different age groups. Eye tracking studies of users reading a long passage on a scrolling interface found that saccade amplitudes were the strongest predictor for age groups (Shojaeizadeh & Djamasbi, 2018). Saccades are the rapid eye movements between each fixation. Based on the regression analysis older adults make larger forward and backward jumps and make fewer total forward movements, but comprehension does not differ significantly. This indicates that older readers do not necessarily read worse but employ different strategies in order to compensate for their cognitive and visual constraints.

The studies of oculomotor characteristics using eye tracking technology have guided Gaze-contingent reading interfaces today, highlighting the fundamental constraints and mechanisms of visual processing during reading (Medan & Pelman, 2024). These interfaces leverage our understanding of perceptual spans, lexical capacities, and the relationships between eye movements and cognitive processing in order to develop adaptive reading environments that enhance performance and reduce cognitive load for readers (Medan & Pelman, 2024). Practical implementations of gaze-contingent reading interfaces translate these cognitive principles into specific visual techniques. A prime example is the "New Line" (NL) interface, which employs several gaze-responsive strategies (Medan & Pelman, 2024). The interface employed several gaze-responsive strategies such as a gaze-driven auto scrolling mechanism where text scrolls horizontally and scroll speed is dependent on the speed of the user's eye movement across the screen (Medan & Pelman, 2024). This design's goal was to maintain continuous forward reading and remove the need for manual scrolling. It also highlighted the word in the center of the gaze, which aided in reinforcing focal attention and reducing peripheral distractions, mimicking the same principle of using a finger point for guided reading (Medan & Pelman, 2024). The visible character range matches the average perceptual span (Medan & Pelman, 2024). With a gaze activated navigation system where you can look at the edge of the screen to change chapters, essentially enabling hands-free interaction (Medan & Pelman, 2024). This research did not produce a significant difference in comprehension or reading speed but provided insight into the potential of gaze-contingent interfaces to restructure spatial navigation and focal guidance without degrading performance (Medan & Pelman, 2024).

Another class of digital reading intervention restructures the dynamics of reading through a rapid serial display of text known as RSVP or Rapid Serial Visual Presentation (Benedetto, Drai-Zerbib, Pedrotti, & Baccino, 2015). In RSVP systems, words are presented sequentially at a fixed central location, eliminating the need for horizontal saccades and regressions (Benedetto et al., 2015). By removing spatial navigation demands, RSVP reduces regression opportunities and enforces controlled pacing. This intervention has been popularized in support of an alternative form of reading for individuals with ADHD traits, as it minimizes the peripheral distractions and limits gaze dispersion. Empirical findings suggest that RSVP can increase reading speeds under certain conditions and may reduce oculomotor load (Rubin & Turano, 1992; Benedetto et al., 2015). However, comprehension outcomes are mixed, particularly for longer or conceptually complex texts (Acklin & Papesh, 2017). Because readers are not able to re-inspect previous words, higher-level integration and reanalysis processes may be constrained (Acklin & Papesh, 2017). While RSVP addresses spatial instability by removing eye-movement, it does not model graded recovery dynamics during sustained readings.

Recent work has also explored the possibilities of assistive reading in dynamic environments, specifically with handheld devices or smartphones (Lei et al., 2023). DynamicRead employed gaze-based scrolling techniques for mobile reading applications (Lei et al., 2023). DynamicRead evaluated four gaze scrolling interfaces, including gesture-based (Eye-Swipe), dwell-based (Hitbox), pursuit-based (Moving Bar), and implicit reading-speed-based auto-scrolling methods (Lei et al., 2023). Their findings suggest that explicit gaze activation techniques are more robust under dynamic conditions such as walking, whereas implicit auto-scrolling struggled when gaze patterns were irregular (Lei et al., 2023). Notably, the auto-scrolling technique, which relied on predicting reading speed from gaze trajectories, performed poorly when readers exhibited non-linear behaviors such as fixations, regressions, or attention lapses (Lei et al., 2023). These findings highlight the challenge of modeling natural reading variability and suggest that simple regression-based prediction mechanisms may be insufficient for supporting sustained reading continuity (Lei et al., 2023).

Aside from guided reading interventions using auto-scrolling and restructuring approaches, adaptive typography systems have also been explored as a means of supporting digital reading performance (Chen et al., 2026). One such example is SITUFont, which dynamically adjusts typographic parameters such as spacing and font characteristics to optimize readability in mobile environments (Chen et al., 2026). Unlike gaze-contingent systems, SITUFont adapts to environmental and situational factors rather than real-time oculomotor behavior (Chen et al., 2026). While adaptive spacing and layout modifications may improve readability under certain conditions, empirical findings indicate that such interventions do not consistently yield significant improvements in comprehension outcomes for older adults (Chen et al., 2026). These findings suggest that while visual parameter optimization can mitigate perceptual constraints such as crowding and contrast sensitivity decline, typography-based adaptation alone may be insufficient to address the dynamic recovery and re-orientation challenges observed during sustained reading (Chen et al., 2026).

Research has also identified several other experimentally validated gaze-contingent techniques designed to guide attention and stabilize reading (Rummens & Beier, 2025; McConkie & Rayner, 1975). These include dynamic word highlighting, where the fixated word is emphasized in real time (Rummens & Beier, 2025); focus-window masking, which restricts visible text to a region aligned with the perceptual span (McConkie & Rayner, 1975); and line guides that reduce navigation errors during regressions. An example of focus-window masking is the moving window paradigm, in which text outside a predefined region or perceptual span of the reader is masked or replaced in real time using gaze inputs (McConkie & Rayner, 1975). The window size is predefined and experimentally manipulated with masking updated continuously as the eyes move, allowing for the entire system to run dynamically during natural reading (McConkie & Rayner, 1975). Experimental findings show that reading speed decreases when the window size becomes too small the reader's parafoveal preview of upcoming words gets eliminated, limiting the reader's ability to anticipate the lexical content of succeeding words (McConkie & Rayner, 1975). As a result, fixation durations increase and regressions become more frequent, leading to measurable declines in overall reading rate (McConkie & Rayner, 1975). This highlights that gaze-contingent manipulation of visible text directly influences reading efficiency and proper considerations are required to shape a viable interface mechanism.

One recent implementation of a gaze-based highlighting system was demonstrated in a controlled classroom setting (Rummens & Beier, 2025). It used eye tracking in order to detect the fixated word of the user and then changes the word dynamically from black to blue (Rummens & Beier, 2025). This design aims to mimic finger-point reading and operates continuously during natural reading (Rummens & Beier, 2025). 70 second-grade children (8-9 years old) who are primarily native Danish speakers were evaluated upon a number of metrics such as reading speed, fixation duration, number of regressions, and comprehension (Rummens & Beier, 2025). From the results, the system produced faster reading speeds alongside shorter fixation durations with no negative impact on comprehension (Rummens & Beier, 2025). This study was able to demonstrate the feasibility of gaze-based interface systems in real classroom environments as well as performance improvements without comprehension tradeoffs (Rummens & Beier, 2025).

Beyond general highlighting techniques, other assistive systems have been developed to target specific reading breakdown behaviors (Wang et al., 2024). GazePrompt was developed with two adaptive features: line switching support, which detects regressions and highlights the intended line or shows a directional arrow for guidance, as well as a word support system that detects prolonged fixation and triggers magnification or text-to-speech (Wang et al., 2024). This system was developed to aid users with visual impairments, with a total 13 low-vision participants with a follow-up silent reading study was conducted where the time it took to switch between each line was recorded as well as the frequency of word recognition errors and numerous other metrics (Wang et al., 2024). Results showed a significant reduction in line-switching time, with reduced word recognition errors and heightened concentration as well as an improvement in perceived comprehension (Wang et al., 2024). This showcases that gaze-triggered micro-intervention can prove to be a beneficial assistive tool for the visually impaired (Wang et al., 2024).

Collectively, these systems demonstrate the feasibility of gaze-contingent and adaptive reading interfaces across multiple design strategies, including layout restructuring, gaze-driven interaction, adaptive typography, and attention-guiding techniques. However, most approaches focus on improving navigation, reducing oculomotor load, or optimizing visual presentation, rather than modeling the underlying dynamics of reading instability. While some systems introduce adaptive features, they are often applied uniformly or triggered by predefined conditions without accounting for individual variability or sustained reading behavior. As a result, current interfaces lack a unified framework for interpreting gaze behavior as a continuous signal of reading stability, limiting their ability to provide targeted and context-sensitive support during moments of difficulty.

2.3 Personalization, Modeling & Detection Limitations

While the systems discussed in Section 2.2 demonstrate the feasibility of gaze-contingent and adaptive interfaces, their effectiveness depends fundamentally on how reading difficulty is detected and how the adaptive decisions are modeled. This section examines the triggers and personalization strategies that underlie these systems. This analysis will examine the representative systems in turn, beginning with the interaction-driven gaze-based mechanism.

At the interaction level, DynamicRead adopts a gaze-triggered scrolling mechanism (Lei et al., 2023). Gaze-based actions for scrolling techniques, such as moving gaze vertically to trigger vertical swiping for page advancement (Eye-Swipe), fixations within a designated region near the bottom of the page for a short duration can scroll the page forward (Hitbox), a visual element that is present that prompts readers to follow it with their eyes to activate scrolling (Moving Bar), and a gaze pattern analyzer that predicts a reader's reading pace and auto scrolls for the users (Auto-Scrolling) (Lei et al., 2023). With that being said, DynamicRead adapts at an interactive level and enhances the reader experience, but it does not directly adapt or analyze to cognitive difficulties via eye-movement (Lei et al., 2023). While DynamicRead demonstrates the feasibility of gaze-based interaction under dynamic conditions, its adaptation is limited to input-level control rather than behavioral interpretation. It does not model reading instability or account for cognitive difficulty, highlighting a gap in systems that distinguish between gaze as interaction versus gaze as a signal of reading state.

In contrast to interaction-driven systems, GazePrompt attempts to detect moments of reading difficulty using gaze-derived behavioral thresholds (Wang et al., 2024). Rather than treating eye movements as explicit commands, the system monitors natural reading behavior to infer when a reader is experiencing instability or cognitive strain (Wang et al., 2024). These reading behavior includes prolonged fixation durations, an increase in regression frequency, and an irregular saccadic pattern (Wang et al., 2024). When these thresholds are prompted, GazePrompt automatically triggers adaptive visual prompts to guide the reader to a stable reading forward progression (Wang et al., 2024). This includes providing a subtle highlight of the current line, emphasis on the next word for semantic continuity, and even directional cues (Wang et al., 2024). Furthermore, unlike gesture-based scrolling systems via gaze, GazePrompt is triggered automatically and detects changes in reading behavior (Wang et al., 2024). However, one of the system's limitations is that gaze-behavior may not totally reflect cognitive difficulty, as reading behavior can mean different states, such as prolonged fixations can mean deeper processing, or regressions can mean they are part of the comprehension strategy of readers (Wang et al., 2024). As such, due to the system's predefined behavioral thresholds, it may misinterpret gaze-behavior and may produce false positive results (Wang et al., 2024). Although GazePrompt introduces behavioral detection through gaze-derived thresholds, its reliance on fixed rules limits its ability to generalize across readers and contexts. The absence of baseline-relative modeling and ambiguity in interpreting gaze patterns reveal a need for more robust, personalized, and continuous representations of reading instability.

Moving towards an infrastructure perspective, CalibRead focuses on gaze alignment optimization rather than behavioral instability detection (F. A. Zhang et al., 2024). Eye trackers use cameras to capture images of your eyes and detect pupil positions, eye corners and sometimes head pose in order to determine which point on the screen you are looking at (F. A. Zhang et al., 2024). Now this general mapping typically comes with a few challenges, namely, mapping being sensitive to environmental and geometric variations, such as head pose, viewing distance, device orientation, and lighting conditions (F. A. Zhang et al., 2024). These factors introduce systematic alignment errors that necessitate calibration (F. A. Zhang et al., 2024). Traditional calibrations designed to solve this issue make users look at a dot on the screen whilst the camera captures images of their eyes and stores the features and extracts the exact coordinate of that dot (F. A. Zhang et al., 2024). This process is usually repeated 7 or 9 times, allowing the system to build a dataset mirroring eye patterns and screen position (F. A. Zhang et al., 2024). Now this system requires explicit user participation and interrupts reading, as well as needing recalibrations if drift occurs (F. A. Zhang et al., 2024). CalibRead aims to solve this by designing an unobtrusive and continuous calibration mechanism that operates during natural reading (F. A. Zhang et al., 2024). They approach this problem by taking advantage of the structured layout of text (F. A. Zhang et al., 2024). During readin, gaze points generally fall along horizontal lines where the words are located (F. A. Zhang et al., 2024). When gaze points systematically deviate from the text lines, the system interprets this as calibration drift and updates the gaze-to-screen mapping accordingly (F. A. Zhang et al., 2024). These updates occur incrementally during normal reading, improving alignment without requiring explicit recalibration (F. A. Zhang et al., 2024). With this modeling, the system showed reduced spatial error compared to traditional calibration, as well as improved accuracy without requiring explicit calibration in sustained natural readings (F. A. Zhang et al., 2024). However, the system operates solely at the sensing layer. While improved alignment strengthens the reliability of gaze input, it does not attempt to interpret reading behavior, detect instability, or trigger adaptive support (F. A. Zhang et al., 2024). CalibRead improves the reliability of gaze input through continuous calibration, but remains confined to the sensing layer. While accurate gaze data is necessary for downstream analysis, the system does not extend into behavioral modeling or adaptive decision-making, indicating a gap between measurement precision and interpretability.

On the other hand, SituFont approaches adaptation through contextual environmental modeling (Chen et al., 2026). SituFont's main problem is addressing the Situational Visual Impairments (SVI) with mobile reading (Chen et al., 2026). SituFont aims to develop a model that adapts font size, weight, and spacing to meet situational needs (Chen et al., 2026). It uses smartphone sensors that detect ambient light and motion sensors (Chen et al., 2026). In order to detect environmental contexts and use machine learning in order to predict the proper textual format (Chen et al., 2026). This uses a human-in-the-loop mechanism as well by allowing users to be able to refine the adjustments and systematically personalize them to fit their demands (Chen et al., 2026). Through this, the research was able to see improvement in reading efficiency under simulated SVI conditions and reduced perceived workload with users preferring the adaptive support system over standard static settings (Chen et al., 2026). SituFont demonstrates that adaptive typography can be systematically modeled rather than manually configured (Chen et al., 2026). Showcasing how environmental context can serve as triggers for real-time interface adjustments and how machine learning can be integrated with user feedback to progressively personalize adaptation (Chen et al., 2026). More importantly, it illustrates a shift from static accessibility settings towards more responsive and context-aware reading interfaces (Chen et al., 2026). SituFont demonstrates that adaptive typography can respond to environmental conditions and incorporate user feedback for personalization. However, its adaptation is driven by external context rather than real-time reading behavior, and it does not verify whether the reader is experiencing difficulty. This highlights the limitation of context-driven adaptation in capturing internal cognitive states.

Taken together, these systems demonstrate that adaptation in gaze-based reading interfaces can occur at multiple levels, including interaction-driven, behavior-threshold driven, infrastructure-level sensing, and context-aware modeling. However, instability is often indirectly inferred, treated as a binary threshold event, or not modeled explicitly. Few systems represent reading instability as a continuous, baseline-relative process, and even fewer account for recovery dynamics following disruptions. This reveals a critical gap in the literature: while gaze-based systems have advanced in sensing accuracy and interface design, they remain limited in their ability to model and respond to graded reading instability over time. Addressing this gap requires moving beyond fixed triggers toward continuous, personalized stability modeling capable of supporting adaptive visual scaffolding.

Chapter 3

Theoretical Framework

This chapter establishes the theoretical basis for the proposed system. Rather than surveying empirical findings, which are treated in Chapter 2, it advances an argument in which each theory initiates the next: reading is a cognitive process made observable through eye movements (Section 3.1); aging slows that process and inflates the effort of recovering after a disruption (Section 3.2); this recovery effort constitutes a cognitive load that interface support can offload (Section 3.3); and such support is most effective when it is graded and faded according to need (Section 3.4). These strands converge on a single construct, the reading stability index (Section 3.5). The chapter then turns from why the system should work to how it is realized as an interaction: how gaze is sensed from a consumer webcam (Section 3.6), and how detected instability is translated into visual support through interaction design grounded in human–computer interaction principles (Section 3.7).

3.1 Eye Movements as a Window into Cognition

Reading isn't simply a visual activity but a whole cognitive process in which words are identified and integrated into meaning, and this process unfolds through the movements of the eyes. The theoretical justification for inferring cognition from gaze rests on two complementary positions.

The Eye–Mind Hypothesis (Just & Carpenter, 1980) proposes that during reading, eye movements closely reflect ongoing cognitive processing. Specifically, it states that readers tend to maintain their gaze on a word for as long as it is being processed, allowing fixation location and duration to serve as real-time indicators of attention and processing effort. An example of this would be observing one's eye movement as they navigate a sentence, "The cat sat on the mat" and "The cat sat on the photosynthesis" notice how when your eyes reach "photosynthesis," they often remain there longer. This is because the brain needs additional time to identify the word, retrieve its meaning, and determine how it fits into the sentence. In its strongest form, the hypothesis assumes little to no delay between where the eyes are directed and what the mind is processing. Although this assumption has been questioned, particularly by studies on covert attention, which suggest that cognitive focus can sometimes shift independently of gaze direction, the Eye–Mind Hypothesis remains the dominant theoretical foundation for eye-tracking research in reading—the domain for which it was originally developed and where empirical support is strongest. Consequently, gaze behavior continues to be widely accepted as an observable proxy for otherwise unobservable cognitive processes (Nedaei, Säljö, Kanwal, & Goodchild, 2025). Under this framework, prolonged fixations, repeated fixations, and backward eye movements (regressions) are interpreted as indicators of increased cognitive processing demands. These gaze behaviors provide measurable evidence of moments when readers experience greater effort in word recognition, comprehension, or reorientation within the text, making them valuable indicators of reading difficulty and cognitive workload.

The E-Z Reader model (Reichle, Pollatsek, Fisher, & Rayner, 1998; Reichle, Rayner, & Pollatsek, 2003) supplies the mechanism behind the relationship between gaze and cognition proposed by the Eye-Mind Hypothesis. While the Eye–Mind Hypothesis states that eye movements reflect cognitive processing, the E–Z Reader model describes the process that links the two. It proposes that readers first perform a rapid familiarity check to determine whether a word is recognized. Once this initial recognition occurs, the brain begins preparing the next eye movement while continuing to process the current word. The model also assumes that readers focus their attention on one word at a time rather than processing several words simultaneously. Based on these assumptions, the model explains that eye movements are directly influenced by the difficulty of recognizing words. Familiar words are processed quickly, resulting in shorter fixations or being skipped altogether. In contrast, unfamiliar words require more processing time, leading to longer or repeated fixations and regressions. As a result, differences in eye movement patterns reflect differences in the cognitive effort required to process the text.

The empirical reading parameters documented in Chapter 2.1 are consistent with this theoretical picture. Skilled adult readers exhibit mean fixation durations of approximately 200-250 milliseconds, produce around 2.6 regressions per sentence, and skip 21-27 percent of words, a profile in which gaze behavior tracks lexical difficulty at the word and sentence level (Pagán et al., 2025). Older adults, by contrast, fixate for approximately 263 milliseconds on average compared to 232 milliseconds for younger readers, and produce around 3.5 regressions per sentence (Moreno et al., 2018). These documented differences in gaze behavior between age groups are precisely what the Eye-Mind Hypothesis and E-Z Reader predict when processing is slower and integration more effortful, which is the subject of the following section.

Taken together, these positions license the central premise of this study: that gaze metrics such as fixation duration, fixation count, and regressions can serve as observable indicators of the cognitive state of a reader. This premise is what makes it possible, in principle, to detect reading difficulty from gaze alone. It is important to note, however, that the coupling between gaze and cognition is not strictly one-to-one a single fixation can have more than one cause and this limitation is addressed directly in Section 3.5.

3.2 Cognitive Aging and Reading

Having established that gaze reflects cognitive processing, the framework next explains why this study focuses on middle-aged and older adults. According to Salthouse's Processing-Speed Theory (Salthouse, 1996), much of the age-related decline in cognitive performance is caused by a general slowing of mental processing. As cognitive processing slows, the brain requires more time to complete each mental operation. This delay means that information processed earlier may begin to fade from working memory before it can be integrated with later information, making complex cognitive tasks more difficult. As a result, older adults often require more time to process information and are more vulnerable to disruptions during cognitively demanding tasks such as reading.

Applied to reading, slowed processing implies that older readers require more time to identify and integrate words, which is consistent with the longer fixations and more frequent regressions documented in Chapter 2. The theory's importance to this study lies less in steady-state reading than in the *recovery* phase: when reading is disrupted and the reader must re-establish position and re-engage processing, slowed processing compounds the time and effort that recovery demands. Recovery is thus the point at which age-related slowing is most consequential.

The empirical evidence reviewed in Chapter 2.1 bears this out directly. Older adults produce regression path durations hundreds of milliseconds longer than younger adults when syntactically complex or low-predictability sentences are encountered, and show disproportionately larger slowdowns when contextual prediction fails (Moreno et al., 2018; Pagán et al., 2025). Display conditions compound this cost: condensed text spacing produces 20-30 percent more reading disruptions in older adults than in younger adults, and low luminance conditions slow reading by 14-22 percent relative to younger readers (Li et al., 2019; Paterson et al., 2013). These findings confirm that the recovery burden predicted by processing-speed theory is not merely theoretical; it is observable, measurable, and amplified by the digital reading conditions that the target population routinely encounters.

This reasoning grounds two of the study's commitments theoretically rather than merely empirically. First, it identifies middle-aged and older adults as the population in which recovery cost is most pronounced and therefore most worth supporting. Second, it identifies recovery rather than comprehension outcome alone as the appropriate phenomenon to target and measure.

3.3 Cognitive Load and Recovery

If recovery is the phenomenon of interest, Cognitive Load Theory explains why intervening in it should help. The theory (Sweller, van Merriënboer, & Paas, 1998) holds that working memory is sharply limited in capacity, and that the total load placed on it during a task comprises load intrinsic to the task and extraneous load imposed by the manner in which information is presented. Extraneous load consumes the same limited resources as the task itself but contributes nothing to understanding, and well-designed presentation can reduce it.

This study reframes the recovery process as a source of extraneous load. When reading is disrupted, the effort required to locate one's place and re-establish forward progression draws on limited working-memory resources without itself advancing comprehension. For older readers, in whom slowed processing already lengthens recovery (Section 3.2), this extraneous load is correspondingly heavier.

It follows that an interface which assists the reader in re-orienting after a disruption offloads this extraneous load, freeing working-memory resources for comprehension. Cognitive Load Theory thus provides the mechanism by which intervention can help at all: the value of support lies in reducing the extraneous load that recovery would otherwise impose.

The regression path duration evidence from Chapter 2.1 makes this concrete. When older adults encounter difficulty at a given word or sentence, they spend measurably longer periods of re-reading and re-integrating the affected region before they continue moving forward; time that is consumed not by comprehension itself but by the effort of recovery (Moreno et al., 2018).. This re-reading interval is precisely what Cognitive Load Theory identifies as extraneous load: it draws on limited working-memory resources without advancing understanding. An interface that provides a re-orientation cue at the point of disruption reduces the search and re-engagement cost, offloading the extraneous component and freeing resources for the comprehension that follows.

3.4 Scaffolding and Graded Support

The final strand concerns the *form* that support should take. The concept of scaffolding (Wood, Bruner, & Ross, 1976) describes assistance provided to a learner that enables performance otherwise beyond their independent reach, calibrated to current need and progressively withdrawn or faded as competence is regained. The related notion of the Zone of Proximal Development (Vygotsky, 1978) holds that support is most effective when it targets the gap between what a person can accomplish unaided and what they can accomplish with help.

Applied to reading support, these ideas argue against both static and binary intervention. Support should be proportional to the reader's current difficulty: minimal or absent when reading is stable, stronger when the reader is struggling, and withdrawn as stability returns. Constant support risks becoming an extraneous distraction in its own right, while binary support cannot distinguish a minor wobble from a complete loss of position.

This grounds the study's choice of *graded* and *fading* visual scaffolding multiple levels of support, applied in proportion to detected difficulty and removed as the reader recovers. It is the theoretical home of the four-level intervention mechanism described in the methodology.

3.5 The Reading Stability Index

The four theories established in the preceding sections converge on a shared requirement: the system needs a single, real-time measure of how far a reader has departed from their own stable reading at any given moment. The reading stability index (RSI) is that measure. Reading stability is the condition in which lexical processing proceeds with efficient forward progression; instability is the latent state in which difficulty disrupts that progression and imposes recovery effort. Because this state is not directly observable, the RSI serves as its continuously computed, gaze-derived proxy.

Three properties of the RSI follow directly from the theoretical chain. It must be *continuous* because cognitive load and scaffolding support are both graded a binary signal discards the information needed to determine which level of support is appropriate. It must be *baseline-relative* because processing-speed theory establishes that older adults' typical reading is already slower as a stable individual characteristic; measuring instability against a personal calibration baseline ensures that ordinary reading is not misread as chronic difficulty. And it must be *multi-metric* because no single gaze signal is unambiguous; a long fixation can reflect difficulty or deep engagement equally whereas genuine instability produces a concerted disturbance across fixation, dwell, and regression simultaneously, and convergence across signals resolves that ambiguity.

Recovery latency, the study's primary outcome, follows naturally from this definition: instability is departure from a personal baseline across converging signals, recovery is the RSI's return toward that baseline after a disruption, and recovery latency is the time that return takes. The RSI therefore serves a dual role as both the trigger for scaffolding and the basis of the outcome measure. Its formal definition and computation are developed in Chapter 4.

3.6 Gaze Estimation and Sensing

The preceding section established the Reading Stability Index (RSI) as a continuous, baseline-relative composite of fixation, dwell, and regression signals that the system derives from a reader's eye movements and computes in real time. The present section addresses the prior, enabling question: how does a consumer-grade camera produce those signals in the first place?. This is an important question to ask because the effectiveness of the system depends on the quality of its eye-tracking data. If the gaze data is inaccurate or delayed, the system cannot reliably detect reading instability or provide appropriate assistance. Therefore, this section explains how eye-tracking works, how camera images are converted into gaze positions on the screen, the available eye-tracking technologies, and the practical limitations of using an ordinary camera for this project.

3.6.1 The Gaze Estimation Pipeline

To detect reading instability, the system must conduct gaze estimation, which is the computational process of determining where the user is looking based on images captured by a camera; this whole operation can be subdivided into a four-stage process. In the first stage, a face detection algorithm locates the user's face within each captured frame (usually a CNN like BlazeFace or MTCNN, running in real time), which captures a bounding box identifying the region of the face in the image. In the second stage, now that we have located the face within the image, facial landmark localization identifies the eye regions within the detected face, normalizing it and then producing a small rectangular patch that is standardized for size and orientation; for example, it would crop the image, keeping only the eye region, and then rotate or resize in order to produce an eye patch as output. Normalization is important in this step, as the user's position from the camera can vary greatly, whether they are sitting close to the camera or further, or tilting their head; normalization is used in order to minimize variations unrelated to gaze direction. Afterwards, in the third stage, a regression model, typically a Convolutional Neural Network (CNN), is adapted in order to pinpoint where the user is looking based on the eye patch and from this the model can express to either a three-dimensional gaze direction, expressed as pitch and yaw angles relative to the camera, or a two-dimensional point of gaze (PoG) directly on the screen surface as x,y coordinates. Lastly, on the fourth stage, even after Stage 3, you don't yet have coordinates that mean anything to your reading interface. This is because the gaze model was trained on data from many different people and setups. As it stands it's currently a generic model which needs to be properly calibrated for the users screen, camera position, and eyes. After calibration, the model's raw outputs are transformed through this user-specific mapping before being used as gaze coordinates. The result is considerably more accurate than the raw model output (Krafka et al., 2016).

3.6.2 Sensing Approach and Spatial Accuracy

There are two main approaches for gaze estimation and the choice between them is largely determined by the available hardware (Hansen & Ji, 2010). The first approach, geometric estimation, achieves high accuracy by shining infrared light at the eye and tracking the reflection off the cornea. Research-grade systems using this method report accuracy of 0.25 to 0.5 degrees of visual angle. The limitation is that it requires a dedicated infrared illuminator, which consumer laptop webcams and smartphone front cameras simply do not have. The second approach, appearance-based estimation, learns directly from the pixel appearance of the eye and face using a CNN trained on large labeled datasets, with no need for infrared or any specialized hardware (X. Zhang, Sugano, Fritz, & Bulling, 2015; Cheng, Wang, Bao, & Lu, 2024). Because this project's deployment targets are ordinary RGB cameras, appearance-based estimation is the only viable approach.

The drawback of using appearance-based gaze estimation is that it is less precise than infrared eye tracking. On consumer devices such as webcams, the estimated gaze position is usually off by about two to four degrees of visual angle (Molina-Cantero et al., 2024). At a normal reading distance of around 60 centimeters, this error becomes roughly two to four centimeters on the screen. This is larger than a single letter or short word, meaning the system cannot accurately tell which exact character a person is looking at. Instead, it can reliably determine the text line or general word group being viewed. For this reason, the RSI and the adaptive reading interface operate at the level of text lines or multi-word sections rather than individual characters.

3.6.3 Calibration, Drift, and Frame Rate

Using an ordinary camera for gaze tracking introduces two practical constraints that shape what the RSI can and cannot measure. The first is calibration drift: even after the initial calibration, accuracy degrades over a session as the user shifts posture, moves their head, or the lighting changes. WebGazer.js has been reported to exhibit error growth from approximately five to ten centimeters over twenty minutes without active correction (Papoutsaki, Gokaslan, Tompkin, He, & Huang, 2018). For participants in this age range, repeatedly stopping to recalibrate is disruptive and cognitively taxing, so the system instead anchors gaze features to text-line positions rather than absolute screen coordinates , meaning uniform drift does not corrupt the RSI's relative comparisons. The second constraint is frame rate: consumer cameras run at approximately 30 Hz, producing one sample every 33 milliseconds, compared to 250 to 1000 Hz for research-grade trackers. Fixations last 100 to 800 milliseconds and are captured across multiple samples, so they are reliably detectable; dwell and regressions follow from the fixation record and are likewise measurable. Saccades, at 20 to 80 milliseconds, occupy at most two to three samples, too few to characterize velocity or onset reliably, and microsaccades are undetectable entirely (Valliappan et al., 2020). The RSI is therefore built exclusively from fixation, dwell, and regression signals, which is all the instability detection framework requires.

3.6.4 Gaze Tracking on Android and Available Tools

The primary deployment platform for this project is an Android smartphone, using the front-facing camera as the gaze input device. As established in Section 3.6.2, Android smartphone front cameras are limited to RGB sensors that follow the same four-stage appearance-based pipeline as laptop webcams. What differs from the two is the software ecosystem. Laptop webcams already have several open-source tools while the android native landscape is less mature and underexplored.

The majority of mobile gaze estimation research and applications have been conducted on iOS devices, largely attributed to Apple's ARKit TrueDepth infrared sensor and standardized hardware across a limited device range. This makes accurate gaze estimation significantly more mature than on Android. One of the largest mobile gaze dataset, GazeCapture (Krafka et al., 2016) is exclusively collected on iPhones and iPads; its authors specifically chose iOS over Android to avoid hardware fragmentation. The strongest Android-specific result, 0.46 centimeters on a Google Pixel 2 XL (Valliappan et al., 2020), relied on a proprietary unreleased model and a single flagship device. On unconstrained, uncalibrated Android tablets, TabletGaze (Huang, Veeraraghavan, & Sabharwal, 2017) reports 3.17 centimeters mean error. Android also lacks a hardware-assisted gaze API equivalent to Apple's ARKit TrueDepth sensor. Google's MediaPipe Iris provides real-time iris landmarks on Android but explicitly does not estimate gaze direction (Yuan & Vakunov, 2020). The most relevant open Android system, GAZEL (Park, Park, & Cha, 2021), reports a 27.5 percent relative improvement over prior work, but its pre-trained model was only trained on a single individual, making it unsuitable for a deployable gaze tracker without retraining on a general dataset. The realistic expectation for commodity Android without device-specific tuning is therefore approximately 1.5 to 3 or more centimeters, and no open-source Android-native solution currently provides calibrated screen coordinates out of the box without significant additional engineering work.

Given this landscape, the project adopts a staged approach. The primary prototype uses MediaPipe Face Mesh iris landmarks combined with a polynomial regression fit over a nine-point calibration grid to produce on-screen coordinates. This is fully open-sourced, on-device, and license-free with expected accuracy of approximately 3 to 4 degrees of visual angle. There is, however, a commercial fallback option called Eyedid SDK (VisualCamp, 2024), which is to the best of our knowledge the only Android-native SDK that outputs calibrated screen coordinates with built-in calibration and no external hardware, though its development key operates under a 30-day free trial. Higher-accuracy upgrade paths such as a GazeFollower backend over local WiFi or a on-device iTracker TFLite model are planned fo rthe succeeding stage and are described in Chapter 4. All implementations stem from an identical interface to the primary prototype, so the tracker can be swapped without affecting downstream components.

3.7 Visual Scaffolding and Design Conventions

When designing Visual Scaffolding systems or applications tailored to aiding the user, intentional design choices must be made to ensure that the features only give the needed temporary and adjustable support that the user needs to complete the task.

Along with the concept of Zone of Proximal Development (Vygotsky, 1978), these concepts are the core components of the framework designed to provide engagement and the nuanced support to assist the user through an interface without the extraneous distractions.

However, as with the focus of the study, a discussion must be made to also consider crucial design considerations to supplement the target user demographic. Because of how age-related declines in visual acuity, perceptual spans, processing speed etc. alter how visual stimuli are registered (Salthouse, 1996), our design must calibrate itself to the cognitive and sensory profile of our demographic.

Previous studies such as Tabrizi and Shokripour (2015), Bassiri (2012), and Attarzadeh (2011) have shown that scaffolding is an effective tool to use when teaching English reading and comprehension skills. It supports the learners with analysis, contextualizations in an interactive way. Visual scaffolding is a strategy derived from scaffolding which uses visual cues and strategies to help support learners.

To bridge these educational aspects and our demographic, the framework aims to introduce age calibrated visual scaffolding. At the micro level, the gaze contingent word highlighting (Rummens & Beier, 2025) and dynamic line support (Wang et al., 2024) can provide immediate tracking anchors to mitigate the visual acuity decline and reentry or navigation errors. At the macro level, a study (McConkie & Rayner, 1975) has opted to use an adaptive focus window derived from the moving window paradigm to artificially compress the layout. The goal being to softly mask peripheral noise to accommodate older learners' perceptual spans. Lastly, the temporal pacing is calibrated through adaptive fading. Withdrawing visual supports slowly over time to ensure that the supports last long enough for the increased fixation times and decreased cognitive processing speeds associated with cognitive aging (Salthouse, 1996).

3.7.1 Relation to the Existing Interface

The gaze driven scaffolding layer specified previously is designed as an addition to NewsMead's existing reading interface rather than a replacement for it. The reading surface and article layout already present in NewsMead remain unchanged under the system we propose. In the study version, users cannot change the text size manually. This ensures that the lines of text sit in the same place for accurate gaze tracking. This restriction is only for the study; in the regular version of the app, users can still adjust the text size normally. What the study instead introduces is a single new layer above the app's surface. Where the interface currently changes only in response to deliberate action and applies that change at once or uniformly, to the whole screen, the proposed layer changes in response to continuously monitored behavioral signals and applies its effect to different parts of the paragraph, like a specific word, line or region that was implicated in the disruption present; all the while withdrawing once the reader's gaze behavior returns back to its expected calibrated range.

The reader is never presented with a new screen, mode, or button. The reading screen they already use gains one level of support at a time, and each level fades as the index returns toward baseline. The modules that realize this, the index that drives it, and the four scaffold levels themselves are specified in Section 4.3.

Chapter 4

Methodology

This chapter describes the activities and methods that will be carried out to accomplish the objectives of the research. It is organized around four components: a formative study to validate the system design with the target population before formal evaluation (Section 4.1), the collection of gaze data from the target population under controlled conditions (Section 4.2), the design and implementation of the adaptive system that detects reading instability and delivers graded visual scaffolding (Section 4.3), and the experimental validation of the system's effect on recovery latency relative to a static interface (Section 4.4). The procedures below describe the planned conduct of the study; the full data collection and system evaluation are undertaken in the succeeding stage of the thesis. Ethical considerations relevant to each activity are discussed within the corresponding section, and the supporting documents are provided in Appendix A.

4.1 Formative Study

Before the main evaluation, a formative study will be conducted to evaluate the reading interface with the target population, identify usability concerns, and examine the behavior of the initial AII component weights and scaffold-level thresholds. The formative phase serves as a pre-test of the initial configuration. Its findings may inform limited adjustments before the selected configuration is frozen for the main evaluation, which serves as the post-adjustment validation of the refined system.

4.1.1 Participants and Procedure

In order to test the efficacy of the interface, an initial screening will be conducted via three to five participants from the target population who will take part in the formative phase. Each participant will interact with the adaptive interface individually in a one-on-one session with a researcher present. This session follows the same structure as the main study session so that the conditions will reflect what participants in the main study will experience.

4.1.2 Data Collection and Analysis

The researcher observes each session and records notes on calibration success, scaffolding behavior, as well as any signs of confusion or disruption that is caused by the interface's design. Following the reading task, a brief semi-structured interview is conducted, asking participants what they noticed during reading, and whether the calibration and adaptive interface design were comfortable. Responses are analyzed across participants to identify recurring concerns.

The formative analysis will also examine whether the initial AII configuration produces excessive, delayed, or unnecessary scaffold activations. The researchers will distinguish whether an observed issue is associated with the component weights, scaffold thresholds, smoothing, persistence requirements, participant baseline, or gaze-data quality.

4.1.3 Design Refinement

Findings from the formative phase will inform limited system refinement before the main evaluation begins. Any change to the AII weights, scaffold thresholds, temporal parameters, or visual presentation will be recorded in a design-decision log containing the original value, observed issue, adjusted value, and reason for adjustment. After formative refinement, the selected configuration will be frozen and used consistently throughout the main adaptive-versus-static evaluation. The formative study does not contribute data to the main research phase; its purpose is to produce a refined and user-validated system before formal testing.

4.2 Data Collection

This section describes the participants, materials, and procedures through which gaze data will be gathered to characterize reading behavior in the target population and to establish the per-user baselines on which the system depends.

4.2.1 Participants and Sampling

The study targets adults aged 45 and above, with no upper age limit. The band begins at 45 because this is where the young-adult cohorts that dominate existing reading eye-tracking corpora thin out, leaving the middle years under-represented relative to both those corpora and the 65-and-older groups typical of aging-reading research. It is left open above so that the sample is not truncated at an arbitrary point within the population the system is intended to serve, and so that suitability is determined by screening on vision, reading capability and consent capacity rather than by age itself. Participants are grouped for analysis by reading and device-use profile rather than by age band. Participants will be recruited through purposive sampling from the surrounding community. A target sample of approximately 15 participants is anticipated for the within-subjects design described in Section 4.4.

4.2.2 Screening and Inclusion Criteria

Prospective participants will be screened prior to data collection to ensure that observed effects are attributable to age-related reading dynamics rather than to clinical or sensory confounds. Inclusion requires the following:

- Aged 45 or above, with no upper age limit.
- Capacity to give informed consent, confirmed before enrolment.
- Near visual acuity sufficient for the reading task with habitual correction, confirmed through a short vision check.
- Completion of at least secondary education and basic familiarity with digital devices, established through a background questionnaire.
- Absence of diagnosed neurological disorders, diagnosed ocular pathology affecting reading, or severe visual impairments that would substantially affect reading behavior.

- A calibration meeting the acceptance criterion described in Section 4.3.2, assessed before the reading conditions begin. Participants are enrolled where the calibration falls in the green or amber band; a calibration in the red band is repeated once in full, and a participant whose second attempt is also red is not enrolled.

Screening combines a brief vision test, a short reading task to confirm baseline reading capability, and a demographic and device-usage questionnaire. To protect participants, session durations will be limited to prevent fatigue, participation will be voluntary with informed consent obtained beforehand, and all recorded data will be anonymized and used solely for research purposes.

4.2.3 Reading Materials

Reading stimuli consist of real news articles drawn from the NewsMead platform on which the system is built, preserving the ecological validity of the reading context (Tiongquico et al., 2024). Each article is paired with comprehension questions to verify engagement and to support the comprehension measure described in Section 4.4. To prevent text difficulty from confounding the condition comparison, article-condition pairings are counterbalanced across participants using a Latin square arrangement: each article appears in the adaptive condition for half the participants and in the static condition for the other half, so that text difficulty is distributed evenly across both conditions at the group level. A standardized font size will be applied across all participants to prevent typography from confounding gaze-metric comparisons.

4.2.4 Baseline Calibration

Two participant-specific preparation stages precede the experimental reading conditions. First, the eye tracker's spatial calibration aligns the gaze estimate with screen coordinates. Second, a standardized baseline-reading phase establishes the participant's characteristic reading behavior under normal, unassisted conditions. During this phase, the participant reads a standardized passage while gaze data are recorded, and the per-user mean and standard deviation of each gaze metric are computed. No visual scaffolding is presented during baseline collection. The resulting profile is estimated once per participant, frozen after collection, stored, and reused across every adaptive and static article read during that session rather than being re-estimated for each article. This individual reading profile serves as the reference against which subsequent instability is defined, allowing the system to interpret deviations relative to each participant's normal reading behavior rather than against fixed population values, as required by the theoretical framework in Chapter 3.

4.3 Software Design and Implementation

This section describes the design of the adaptive system, which is organized as a real-time pipeline that transforms raw gaze into adaptive visual support. The system is implemented as an extension of NewsMead, an existing Android news-reading application for older Filipino adults (Tiongquico et al., 2024), which provides a participatory-design-validated reading interface and serves as the reading surface for this study. The contribution of the present study is the gaze-driven adaptive layer added on top of that surface. The build used for the study runs offline from a bundled article set and bypasses the application's login and registration, so no account is created and no participant data leaves the device during a session.

4.3.1 System Architecture

The system follows a real-time pipeline: *gaze capture* → *fixation detection* → *stability index* → *adaptation decision* → *visual scaffolding*, with the reader's response closing the loop. All stages run on-device on the Android phone using the front-facing camera as the gaze input, with no external hardware or network connection required during a session. The reading interface and scaffolding layer are rendered within the NewsMead article view; gaze capture and the stability analysis run as a background service communicating with the presentation layer in real time. This on-device architecture supports both the controlled lab sessions of the summative evaluation and potential future extension to naturalistic reading outside the lab.

The adaptive layer is organized as five groups of modules. A *sensing* group drives the front camera and extracts eye and iris features from each frame. A *mapping* group converts those features into a screen coordinate using the per-participant calibration. An *interpretation* group resolves each coordinate to a line of text, detects fixation, dwell and regression events from the resulting line, and maintains the instability index. A *decision* group applies the scaffold thresholds together. A *presentation* group renders the selected scaffold. Only the last of these draws anything, and it draws onto the article view that NewsMead already provides rather than onto a surface of its own. The article view is used in both directions: it supplies the line geometry that makes line-level attribution possible, and it receives the overlay. Figure 4.1 shows the modules and the paths between them.

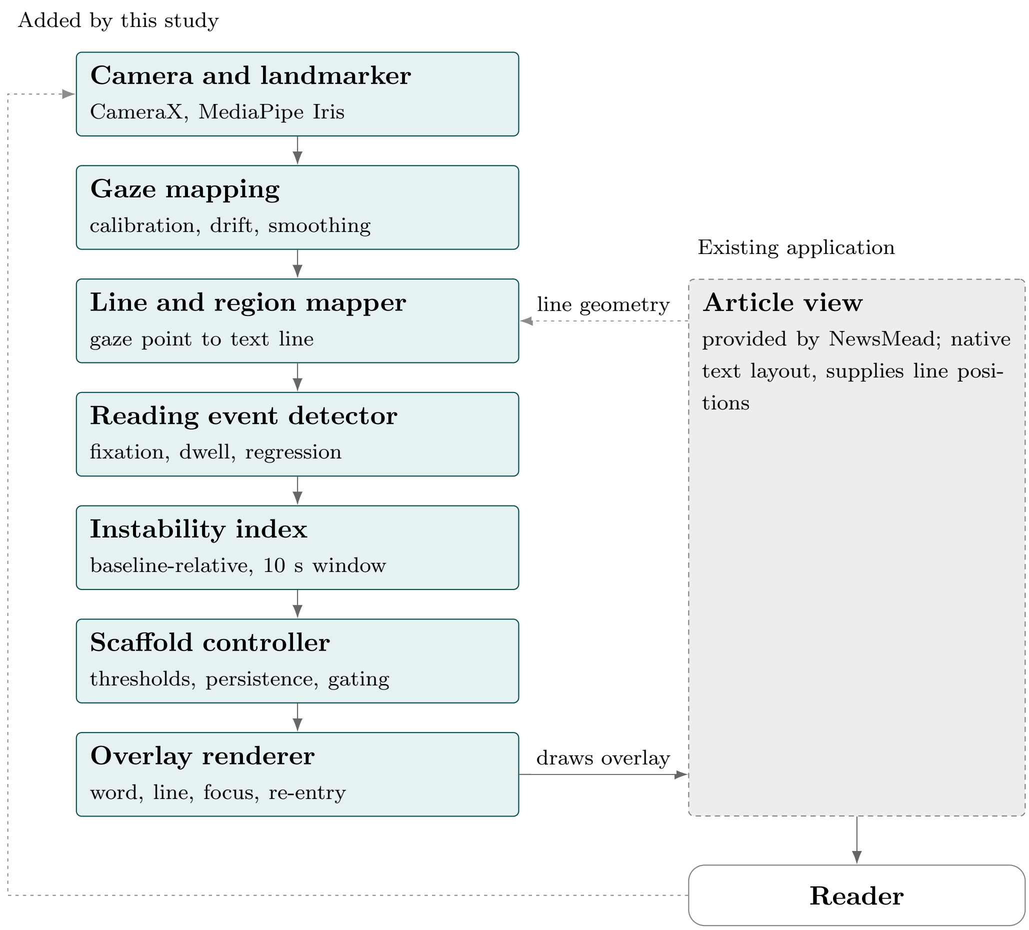

Figure 4.1: System architecture and its interface with the existing application

4.3.2 Gaze Tracking

Gaze-tracking is implemented on the Android front-facing camera using an appearance-based estimation approach, as established in Section 3.6 of the Theoretical Framework. Because consumer-facing Android cameras only provide RGB images without infrared illumination, appearance-based deep learning estimation is the only viable method for this platform, and a per-subject calibration step is required to map model outputs to each participant's screen coordinates.

To select among available on-device and hybrid approaches, six gaze-estimation methods were compared. Three were implemented and measured on the target hardware (Samsung Galaxy A56); the remaining three were assessed from published or vendor-reported figures, either because no Android-native implementation was available or because the method is not deployable on the target platform. Table 4.1 summarizes the comparison, indicating for each method whether its figure was measured in this study or taken from its source.

Accuracy requirement

The requirement for this system is that the line of text a participant is reading can be identified from the gaze estimate. The relevant quantity is therefore *vertical* error, and the relevant unit is the rendered height of one line of body text rather than an absolute distance. Horizontal error displaces the estimate along a line and does not, by itself, cause a line to be misidentified.

Reading text is standardized at 22 dp with a 1.6 line-spacing multiplier, giving a rendered line height of approximately 0.6 cm on the target device. The size is specified in density-independent pixels rather than scale-independent pixels so that it is not altered by the operating system's accessibility font-scale setting; the rendered line height, and therefore the line boundaries used for area-of-interest mapping, are then fixed for a given device regardless of a participant's system settings. The accuracy requirement is accordingly stated as **vertical error below one rendered line height**, approximately 0.6 cm. An earlier formulation of this requirement as a fixed 0.5 cm threshold was derived from a smaller provisional font size and is superseded; the requirement was recomputed against the font actually used, not relaxed to accommodate a result.

Evaluation protocol

Measurements were taken on a Samsung Galaxy A56 held in portrait orientation. Calibration presented sixteen targets and the accuracy test presented a three-by-three grid at positions deliberately offset from the calibration targets, so that the reported error is generalization error rather than residual error of the fit. Error was computed per target from the median of a settled sampling window and is summarized both overall and along the vertical axis alone. The calibration and drift-correction procedures used by the deployed system are described later in this section. Measurements were performed by members of the development team rather than by participants in the target age range, on a single device, at an unrecorded viewing distance and under uncontrolled lighting. They are sufficient to support a comparative selection among candidate methods, which is the use made of them here, but not to establish the accuracy the system will achieve with the study population; that is to be established during pilot testing, and the corresponding limitation is stated in Section 1.3. The individual runs are listed in Appendix C.

Results

Table 4.1: Comparison of Gaze-Estimation Approaches on Android Devices

Approach	On Device	Accuracy	Fps	Calib.	Verdict
GazeFollower	No: laptop + WiFi	0.55 cm[m]	30	16 pt	Most accurate measured, but rig-bounded
MediaPipe Iris + Poly	Yes	1.68–2.42 cm; vertical 0.60–1.01 cm[m]	∼30	16 pt	Selected; adequate vertically under controlled conditions
GAZEL (retrained)	Yes	∼3.9 cm[m]	—	5 pt per-user	Insufficient
iTracker / GazeCapture	Yes	2.46 cm[r]	∼15	per-user	Would need building and training
GazeRecorder	No: browser and cloud	$< 1°$, desktop only[r]	—	30 pt	No SDK
WebGazer.js	No: browser	∼4 cm[r]	—	clicks and cursor	Low accuracy

[m] Measured in this study on the Samsung Galaxy A56. [r] Reported figure: iTracker (Krafka et al., 2016), WebGazer (Papoutsaki et al., 2016), GazeRecorder vendor-published.

Only the selected method was evaluated with error resolved by axis; the GazeFollower and GAZEL figures are combined Euclidean error, because at the time each was assessed the question was whether the approach was viable at all rather than how its error was distributed. The comparison is therefore conservative toward the two rejected methods, which would both appear somewhat better if resolved by axis in the same way. GazeRecorder is left in degrees of visual angle because it is a desktop measurement at a desktop viewing distance; converting it to a linear error at handheld reading distance would misrepresent the condition under which it was obtained.

GazeFollower was the most accurate method measured in this project and was not rejected on accuracy grounds. The GAZEL figure carries a further qualification: it was obtained on held-out subjects from the TabletGaze corpus, recorded on a tablet held in landscape orientation rather than on the target phone in portrait, and the corpus subjects are aged 20 to 40 while this study targets readers aged 45 and above. The retraining was carried out across progressively larger subject sets, and the accuracy plateaued: doubling the number of training subjects improved it by approximately nine percent, while exhausting the remainder of the corpus improved it by a further half percent. A separate pass addressing overfitting made the calibrated result slightly worse rather than better. Both obvious remedies for an underperforming model were therefore attempted and shown not to help, so the figure represents a floor for the approach rather than an artefact of insufficient training data; the progression is tabulated in Appendix C.

Selection rationale

This study adopts MediaPipe Face Landmarker (Iris) with a calibrated second-degree polynomial regression. It is the only candidate that is simultaneously fully on-device, license-free, deployable without a co-located laptop, and free of any requirement to train a model.

GazeFollower was the more accurate method and was nonetheless set aside. It requires a laptop running a Python backend and a live wireless bridge maintained for the duration of every session. For a study conducted with older-adult participants in community settings rather than in a laboratory, this imposes a setup that must be transported, configured and kept working at each visit, and it introduces a network dependency whose failure ends a session that cannot easily be rescheduled. The approach was implemented, integrated, and validated end to end on the target device before being withdrawn on these grounds. This was a deliberate trade of measured accuracy against deployability, and it is not presented as an accuracy improvement; the accuracy cost is stated in Table 4.1 and carried through the limitations that follow.

The selected method does not straightforwardly satisfy the accuracy requirement, and the remainder of this section states the gap and the grounds on which the design proceeds despite it.

Measured vertical error ranges from 0.60 cm to 1.01 cm across runs, against a requirement of approximately 0.6 cm. The better runs meet the requirement; typical runs exceed it by a factor of roughly 1.1 to 1.7. Overall error is substantially larger, from 1.68 cm to 2.42 cm, but this figure is dominated by a systematic horizontal displacement that does not affect line identification and therefore overstates the difficulty of the task the system actually performs.

Four properties of the design reduce the effective error below the per-sample figure. First, adaptation is driven by an index computed over a ten-second rolling window with two-second exponential smoothing at approximately 30 samples per second, so several hundred samples contribute to each decision and the random component of the error averages down; the per-sample figure is not the quantity driving adaptation. Second, the index is computed from positive deviations against each participant's own baseline, so a systematic offset that is stable within a session largely cancels rather than accumulating; the requirement is a stable error, not a small one. Third, gaze features are anchored to text-line positions rather than absolute screen coordinates, as established in Section 4.3. Fourth, the reading font is standardized at 22 dp with generous line spacing, which improves the ratio of error to line height directly and is independently justified by the accessibility rationale of the study.

The engineering record identifies run-to-run instability, rather than error magnitude, as the binding constraint: repeated measurements on the same setup minutes apart varied by a factor of approximately four, attributable to head-pose drift. This is addressed by the calibration subsystem rather than by the estimator. Calibration now applies per-point outlier rejection, excludes points that fail repeated collection from the fit, and reports leave-one-out residuals together with a five-point held-out validation pass before the session proceeds, so that a poor calibration is identified before reading begins rather than inferred from the data afterwards. A nine-point accuracy check during the session additionally fits an affine correction for posture drift. These measures address calibrations that would previously have been accepted without detection; they reduce the incidence of unstable sessions but do not eliminate the underlying sensitivity of single-camera appearance-based estimation to head pose on a handheld device.

Accordingly, line-level attribution is treated as a claim to be verified in pilot testing rather than as an established property of the system. Level 1 of the scaffolding hierarchy is defined as highlighting of the current word *region* rather than of an individually resolved word, since the horizontal precision required to isolate a single word is not supported by the measured error. Should pilot testing fail to demonstrate reliable region-level or line-level localization, the intervention hierarchy will be revised toward the broader focus-window scaffold already defined at Level 3, and the finer-grained claims will be reported as provisional. The corresponding limitation on spatial granularity is stated in Section 1.3.

Terminology

Three procedures are used by the deployed system and are distinguished throughout. Two of them present fixation targets, but only one estimates a mapping, and they are not interchangeable.

Table 4.2: Distinction between calibration, accuracy verification, and drift correction

Procedure	When	Model estimated
Calibration	16 points, once per participant per session	Second-degree polynomial, six coefficients per axis
Accuracy verification	9 points, at any time	None; measurement only
Drift correction	Fitted from an accuracy verification run	Affine, three coefficients per axis

The nine-point procedure is not a calibration. It never re-estimates the polynomial mapping; it measures error and may fit a corrective affine transform on top of the existing calibration.

Calibration procedure

Calibration presents sixteen targets in a four-by-four grid, inset ten percent from the screen edges. At each target the participant fixates while gaze features are collected for 1000 ms, requiring a minimum of ten valid samples; the componentwise median of those samples is retained as the observation for that target. A target that yields fewer than ten valid samples is re-presented. A target that fails collection twice is marked excluded in the session log and omitted from the fit rather than contributed as a low-quality observation. At least twelve of the sixteen targets must be retained for the fit to proceed.

The mapping from gaze features to screen coordinates is a second-degree polynomial fitted separately for each axis by least squares over the retained targets, using standardized features and ridge regularization with $\lambda = 1.0$ applied to all terms except the intercept. Regularization was introduced to contain overshoot between sparsely sampled targets, which in an earlier unregularized implementation produced estimates several thousand pixels outside the screen for in-range inputs.

Live gaze features falling outside the calibrated feature range are extrapolated along the boundary gradient, capped at 1.5 standard deviations. This replaced an earlier hard clamp, which froze the estimate at a fixed screen position whenever features drifted out of range and thereby produced a stationary estimate that could be mistaken for a stable fixation.

Calibration quality assurance

Immediately after the final target is collected, the fitted mapping is evaluated in two ways before the session is permitted to proceed.

Leave-one-out residuals are computed by refitting the mapping once per retained target, each time excluding that target and measuring the error of the resulting mapping at the excluded position. This requires no additional participant time. The median and 95th percentile of these residuals are reported.

A validation pass then presents five further targets, at the screen centre and the four quadrant midpoints. None of these positions coincides with a calibration target. The error of the fitted mapping at these positions is the reportable per-session accuracy figure.

Both quantities are expressed in line heights as well as pixels, line height being the unit relevant to identifying which line of text is being read. Results are presented to the researcher as a banded recommendation: green at or below 3.0 line heights, amber at or below 4.5 line heights, and red above that, banded on the worse of the leave-one-out and validation medians. A drift flag is raised if head position shifted by more than one line height between the two halves of the calibration.

These bands are anchored to the documented accuracy of the tracker rather than to an ideal figure; they identify a calibration that is unusually poor for this equipment, and a green result is not a claim of line-level precision. The researcher may accept the calibration, re-present the three worst targets, or repeat the procedure in full. The calibration file is written only on acceptance. Participants are enrolled at the amber band or better, as stated in Section 4.2.

Accuracy verification and drift correction

A nine-point verification may be run at any point in a session. It presents a three-by-three grid at 20, 50 and 80 percent of each screen axis. These positions are deliberately offset from the 10, 50 and 90 percent positions used for calibration, so that the measured error is generalization error rather than residual error of the fit.

At each target the first 800 ms are discarded to allow the estimate to settle, the following 1200 ms of smoothed estimates are collected, and the median of those samples is taken as the estimate for that target. Error is the distance between that median estimate and the true target position, converted from pixels to centimetres using the reported display metrics. Per-target errors are logged individually; it was this per-target logging that identified error concentration at the lower edge of the screen in early runs, which a single aggregate figure had concealed.

The nine resulting pairs of predicted and true positions are also sufficient to fit an affine correction of the form truth $\approx A\,\text{predicted} + b$ by least squares. Posture drift within a session is predominantly a change in offset and gain rather than a change in the shape of the polynomial mapping, so an affine correction addresses it without repeating the full calibration. The correction is offered to the researcher together with the measured pre-correction error and the estimated post-correction error, and when applied is stored separately and applied after the polynomial mapping, leaving the original calibration intact and reversible.

4.3.3 Reading Surface and Area-of-Interest Mapping

The reading interface uses NewsMead's existing article view, which renders article content natively as a single Android text view. Each text line is registered as an Area of Interest (AOI). Since the body is processed natively, Android's layout API is able to provide the position of every line of text. For each gaze sample taken from the user the pixel coordinates are translated into the relative line of text at that specific location on the screen. It then uses the gaze's vertical position to determine which line the user is looking at instead of being stored as fixed bounding boxes.

4.3.4 The Session Reading Stability Index

The Session Reading State Index (RSI) is computed in real time from the detected regression, dwell, and fixation behavior recorded during the current article-reading session. With that being said, the components are normalized to bounded values and combined into a score ranging from 0 to 100. Session RSI provides a continuous and descriptive summary of gaze-derived reading behavior across the entire reading session. Additionally, RSI is not normalized against the participant's calibration baseline and does not directly drive the adaptation decision.

Higher Session RSI values indicate that the recorded gaze behavior represents behavior associated with instability such as rereading, hesitation, or increased processing effort. Lower values on the other hand indicate vice versa, signifying that the reader's gaze behavior associated with instability accumulated fewer during the reading session. Sequentially, session RSI is used as a descripive outcome, and not as a direct indicator for selecting what visual scaffold the interface will adapt to.

Since RSI score accumulates throughout the reading session, early instability may keep the score elevated longer after recovery, while later difficulty may be diluted by earlier stable reading. This is why RSI cannot be used as a direct indicator for what visual scaffolding support is needed since it is a representation of the reader's overall session history rather than their moment-to-moment reading behavior. The participant-relative measure used to control real-time visual scaffolding is discussed in the next section. With that being said, the reading stability index score is computed as follows:

$$\text{Session RSI} = 100\,(0.40R + 0.35D + 0.25F)\,,$$

Where R is the normalized regression component, D is the normalized dwell component, and F is the normalized fixation component, Regression has the highest weight across all components as returning from a later line to the current line is mostly a strong indicator of rereading, which reflects plausible indicators of confusion, missed information, or difficulty processing the text. Dwell, on the other hand, is assigned the second-highest weight, as prolonged attention or tracking of a reading line can indicate hesitation or increased processing effort, although it may be an indicator of careful reading or processing of information. Lastly, fixation frequency has the lowest weight as it can be ambiguous to readers, as this can provide useful evidence of effortful reading, but can also be an indicator of reading style. Overall, the weights are assigned based on how directly each gaze metric reflects reading instability, with regression being treated as the strongest indicator, with dwell being a secondary signal of increased processing effort, and fixation duration being a useful metric but actually depending on the reader's reading style.

The values of 0.40, 0.35, and 0.25 are treated as initial theory-informed weights. Prior literature supports the expected ordering of regression, dwell, and fixation but does not prescribe their exact numerical proportions. The initial configuration will therefore be examined during the formative phase and retained or adjusted only when the observed system behavior provides sufficient reason for modification.

4.3.5 The Adaptive Instability Index

Since the Session Reading Stability Index is a decriptive measure of gaze behavior across the entire reading session, the Adaptive Instability Index is a recent, participant-relative measure used for real-time adaptation and scaffold selection. Essentially, it is calculated from regression rate, dwell-time proportion, and fixation rate within a rolling 10-second window. Each recent metric within that rolling window is standardized against the participant's baseline, and only positive deviations associated with their reading effort are considered and retained.

For each metric i, the positive standardized deviation is calculated as follows:

$$z_i^+ = \max\left(0, \frac{x_{i,\text{recent}} - \mu_{i,\text{baseline}}}{\max\left(\sigma_{i,\text{baseline}}, \sigma_{i,\text{floor}}\right)}\right).$$

The raw Adaptive Instability Index is then computed as:

$$\text{AII}_{\text{raw}} = \text{clamp}_{[0,4]}\left(0.40z_R^+ + 0.35z_D^+ + 0.25z_F^+\right).$$

In order to prevent small changes in reader's baseline from producing unsually high AII values, the system uses a minimum standard-deviation values of 1.0 regression per minute, 0.05 for dwell time proportion, and 5.0 fixations per minute. The raw index of is exponentially smoothed using two seconds so that brief changes in gaze behavior does not trigger immediate visual support.

With that being said, higher AII values indicates that the participant's recent gaze behavior within that rolling 10 second window is more unstable relative to their baseline or usual reading pattern. Sequentially, a lower AII is an indicator that that their reading behavior remains closer to the their normal baseline.

The component weights and scaffold-level thresholds are separate parameters. The weights determine the contribution of each gaze indicator to the AII, whereas the thresholds determine which level of visual support is activated. The scaffold mapping below represents the initial configuration to be examined during the formative phase.

4.3.6 Adaptation and Graded Visual Scaffolding

The adaptation decision maps the smoothed Adaptive Instability Index onto four graded levels of visual scaffolding. Each level is applied only when instability is detected and remains elevated for the required persistence interval. Each visual scaffold increases one level at a time, beginning with the least intrusive visual scaffold and progressing to the most intrusive. The scaffold level is selected as follows:

Table 4.3: Adaptive Instability Index scaffold mapping

Adaptive Instability Index	**Scaffold level**
Less than 1.0	No scaffold
1.0 to less than 1.5	Level 1: Word-region highlighting
1.5 to less than 2.0	Level 2: Line emphasis
2.0 to less than 2.5	Level 3: Focus window
At least 2.5 with loss-of-position evidence	Level 4: Re-entry support
At least 2.5 without loss-of-position evidence	Level 3: Focus window

Each intervention is grounded in prior work:

- **Level 1 Word-region highlighting:** the fixated word region is dynamically emphasized, following the gaze-contingent highlighting validated by Rummens and Beier (2025). The emphasis uses a colour already present in the interface; nothing moves, and no other property of the text changes.
- **Level 2 Line emphasis:** the current line of reading is highlighted to reduce navigation errors, similar to the line-support augmentation of GazePrompt (Wang et al., 2024). A soft band is placed behind the line, so that the reader can recover their place without searching for it.
- **Level 3 Focus window:** text outside a region around the reader's gaze is softly de-emphasized. The window is sized to exceed the reader's perceptual span, since a window narrower than the span is known to slow reading and increase regressions (McConkie & Rayner, 1975; Rayner et al., 2009). The

remainder of the paragraph is dimmed rather than hidden, so that its shape, length and position remain available for orientation.

- **Level 4 Re-entry support:** on a detected loss of reading position, a directional cue or line highlight guides the reader back to the correct location, following GazePrompt's line-switching support (Wang et al., 2024). The Level 2 band is retained on the line the reader was last reading steadily, and a directional glyph is placed near the current gaze position, pointing back toward it.

Adaptation is conservative, wherein it starts only after the participant's baseline has been established and at least 65% of the recent gaze points fall within the article text. Supporting visual scaffolds increases when signs of instability continue for 1.5 seconds and withdrawn 3.5 seconds of recovery, increasing each visual scaffold one level at a time. Similarly, level 4 is used only when the reader looks away from the text for at least 800 milliseconds and returns at least two lines away from the previous line, indicating a loss of reading position.

Support is withdrawn as the index returns toward baseline, reflecting the fading principle of scaffolding established in Chapter 3. To reduce the risk of distraction, to which older readers are particularly susceptible, a single intervention is active at a time, the least intrusive support is applied first, and triggering is conservative.

4.3.7 Session Logging and Data Schema

Every quantity described in this proposed schema will be computed on-device during a session. The sensitivity analysis in Section 4.4 requires that recorded data be reprocessed under alternative component weights, which is possible only if the per-window component metrics and the baseline they were standardized against are both recoverable. The formative analysis in Section 4.1 undertakes to distinguish whether an observed problem originates in the weights, the thresholds, the smoothing, the persistence requirements, the participant baseline, or gaze-data quality; those six causes are separable only from records taken below the level of the smoothed index. Recovery latency, as defined in Section 4.4, is a difference between two timestamps, both of which must be recorded.

The guiding principle is to record at the lowest level from which everything above it can be re-derived. Logging only the smoothed index would fix the parameter choices that the formative phase is explicitly intended to leave open. Seven categories will be retained, and Figure 4.2 shows the proposed processing stage from which each category will be recorded.

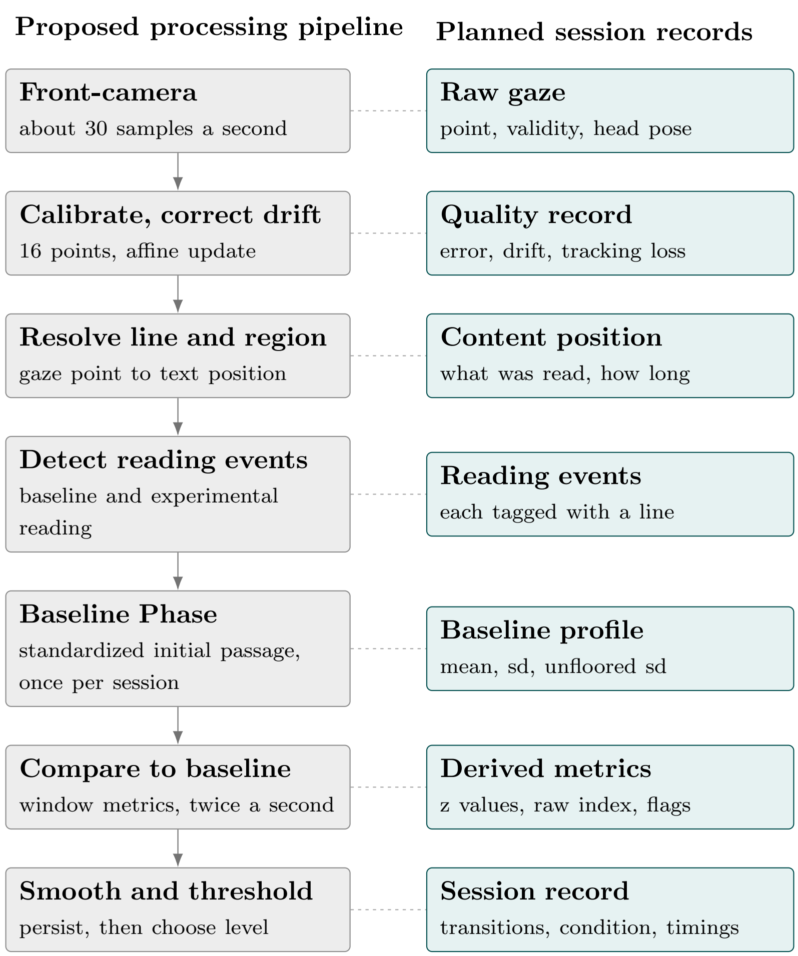

Figure 4.2: Session logging schema mapped to the processing pipeline

Two entries in the schema warrant comment. The system applies minimum standard-deviation values, as described in Section 4.3.4, so that a participant whose baseline reading is unusually consistent does not produce inflated index values. When that floor binds, the index is no longer scaled by the participant's own variability, and the measure is participant-relative in name only for that session. Recording the unfloored standard deviation alongside the floored one, together with a per-window flag, is what allows this condition to be identified rather than inferred. Separately, the proportion of samples falling on visible article text is recorded as a quality measure but is not an accuracy measure: a gaze estimate that lands on the wrong line still counts as valid. It gates adaptation and bounds interpretation; it does not establish that the correct line was identified.

Table 4.4: Session logging schema

Category	**Recording frequency**	**Contents**
Raw gaze	~30 samples per second (30 Hz)	Timestamp, gaze feature vector, mapped screen coordinates before and after drift correction, sample validity, head-pose proxy
Baseline profile	Once per participant	Mean and standard deviation of each metric, the unfloored standard deviation alongside the floored value actually used, baseline passage identifiers and duration
Derived metrics	Twice per second (2 Hz)	Recent regression rate, dwell ratio and fixation rate; the three positive standardized deviations; raw and smoothed index; clamp-hit and floor-bound flags; cumulative Session RSI; active scaffold level
Reading events	Per event	Fixation and dwell events with line index and duration; regressions with source and target line; line changes; off-text intervals; scroll events
Content position	Per event	Article identifier, line and paragraph index over time, cumulative words read
Session record	Per event	Condition and order, phase boundaries, scaffold transitions with level and triggering index value, escalation and withdrawal timings, comprehension responses and response times, and disruption onsets where induced disruptions are used
Quality record	Per session	Calibration leave-one-out and validation medians with the assigned band, drift checks and any affine correction applied, dropped-frame count, proportion of tracking loss, and the proportion of recent samples falling on visible article text

The baseline profile is recorded once per participant and reused across every article read in that session, rather than being re-derived per article. A profile estimated separately for each article would be estimated partly from the reading it is meant to be compared against, and index values from different articles would not be on a common scale.

Records will be written to application-private storage on the device as the session proceeds, so that an interrupted session retains everything collected up to the point of interruption, and will be exported after the session for analysis. Files will be identified by participant code only and will contain no direct identifiers. No camera frames will be retained at any point, consistent with the disclosure in the informed consent form; the raw gaze category will record derived feature values and screen coordinates, not images. At approximately 30 samples per second, a session of the length described in Section 4.4 is expected to produce on the order of a few megabytes per participant, so retention is not expected to be constrained by storage.

4.4 Validation

This section describes how the system's effect on recovery latency will be evaluated, including the experimental design, the participants and procedure, the measures collected, and the statistical analysis.

4.4.1 Experimental Design

System effectiveness will be evaluated using a within-subjects design in which each participant reads under two conditions: the adaptive interface, with the stability index driving graded scaffolding, and a static interface, comprising an identical reading surface without adaptive support. A within-subjects design controls for individual differences in reading ability, which are substantial in this population. To control for learning and fatigue effects, the order of conditions will be counterbalanced across participants, and texts will be assigned so that condition is not confounded with text difficulty.

The counterbalancing is as follows in Table 4.5. Participants are assigned in approximately equal numbers to four groups, which together balance three factors: the order in which the adaptive and static conditions are presented, the pairing of each article with each condition, and the order in which the articles are read. Each article therefore appears in the adaptive condition for half the participants and in the static condition for the other half, so that text difficulty is distributed evenly across both conditions at the group level.

Table 4.5: Counterbalancing of Condition and Article Order

Participant Group	**First Task**	**Second Task**
Group 1	Adaptive (Article A)	Static (Article B)
Group 2	Static (Article B)	Adaptive (Article A)
Group 3	Adaptive (Article B)	Static (Article A)
Group 4	Static (Article A)	Adaptive (Article B)

4.4.2 Participant Selection and Orientation

Participants are drawn from the target population and screened as described in Section 4.2. Prior to the experiment, participants will be oriented to the study through a validation protocol explaining its purpose, procedure, and their right to withdraw at any time without penalty. Informed consent will be obtained before any data is collected; the Informed Consent Form is provided in Appendix A. The pre-experiment vision check, reading task, and background questionnaire serve both as screening and as orientation to the reading setup.

4.4.3 Experiment Procedure

The main evaluation will use the AII weights, scaffold thresholds, and temporal parameters frozen after the formative phase.

Sessions are conducted one-on-one in a controlled setting. After spatial gaze calibration, each participant completes a standardized, unassisted baseline-reading phase in which no visual scaffolding is presented. The resulting participant-specific baseline is then frozen and reused while the participant reads under both the adaptive and static conditions in counterbalanced order, with comprehension questions following each text. Total session duration will be limited to prevent fatigue, with a short break between conditions. During reading, the researcher monitors the session and records structured observation notes on visible signs of difficulty, spontaneous reactions to scaffolding, and any technical issues, without intervening in the reading itself.

4.4.4 Measures and Recovery Latency

The primary outcome is recovery latency. For a disruption event occurring at time t_d, the recovery time t_r is defined as the first moment after t_d at which the stability index returns below the baseline band and remains there for a brief confirmation interval; recovery latency is the difference $t_r - t_d$. This operationalization follows directly from the stability index: recovery is the return of the index to baseline, and recovery latency is the time that return takes. Recovery latency will be compared between the adaptive and static conditions. Secondary measures include reading comprehension (from post-reading questions), reading speed in words per minute, and gaze-derived measures such as regression frequency and fixation duration; these contextualize the primary outcome and verify that any reduction in recovery latency does not come at the expense of comprehension.

4.4.5 Debriefing

The closing phase of each session follows a fixed order: the usability survey is administered first, then the semi-structured interview, and only then the structured debriefing. This is intentioonal so that the debriefing explains what the study measures and what condition causes the adaptive interface to change its display. Providing this information before the interview could influence the participants' recollections and interpretation of their reading experience which could provide bias to the qualitative findings. Conducting the interview beforehand ensures that the observations from each participant is done independently without the influence from the researcher's explanation. The debriefing is delivered from a written script, reproduced in Appendix A, so that it is administered consistently by all four researchers.

After completing both reading conditions, each participant takes part in a semi-structured interview with the researcher. The interview asks what visual changes they noticed during reading, whether anything helped or distracted them, which condition felt more comfortable, and whether they felt the interface had any effect on their reading. Sessions are audio-recorded with participants' consent and transcribed afterward. Two researchers independently code the transcripts and reconcile their themes. The qualitative findings are used alongside the quantitative results to determine whether any observed reduction in recovery latency was experienced and meaningfully attributed to the scaffolding by participants, and to surface usability concerns that gaze metrics alone would not capture. Observation notes recorded by the researcher during the session provide additional context.

Once the interview is complete, the researcher delivers the debriefing. Participants consent under limited disclosure: the consent form states that the reading application adjusts its display and that two versions are compared, but does not state when or why the display changes, and does not state that the adaptive condition responds to detected reading instability. The study title printed at the head of the form refers to recovery cost in general academic terms; it neither defines recovery latency as the measured outcome nor reveals what triggers a change in the display. This is withheld because informing participants in advance that the system reacts to signs of losing their place would lead them to monitor that experience deliberately, altering the behaviour the study measures. Nothing false is disclosed at any point; the camera use, eye-movement tracking, reading and comprehension tasks, interface conditions, session length, risks, and data handling are all stated accurately before consent. The debriefing closes this gap in full, explaining what was measured, what the adaptive interface was responding to, and why this was not stated at the outset. Each participant is then asked whether they still consent to the use of their data, and may withdraw it at that point without penalty and without giving a reason; the response is recorded for every participant. Participants who withdraw mid-session, or who are excluded after enrolment, are debriefed on the same terms.

4.4.6 Statistical Analysis

Because each participant is measured under both conditions, recovery latency will be analyzed as a paired comparison. The condition differences will first be tested for normality using the Shapiro-Wilk test; if the assumption of normality holds, a paired-samples t-test will be used to compare recovery latency between the adaptive and static conditions, and otherwise the Wilcoxon signed-rank test will be used. A significance level of $\alpha = 0.05$ will be applied, and an effect size will be reported alongside the test statistic. Secondary measures will be analyzed using the same paired procedure as appropriate to their distributions.

A post-study sensitivity analysis will examine whether reasonable changes to the component weights alter the study's conclusions. The recorded gaze data will be reprocessed using the initial weights, equal weights, and selected nearby alternatives. Changes in AII classifications, scaffold activations, and recovery-latency results will be compared. This analysis will be used to determine whether the initial weights are sufficiently robust and to recommend possible adjustments for future work.

Appendix A

Research Ethics Documents

RESEARCH ETHICS CLEARANCE FORM
For Thesis Proposals[1]

Name of student researcher :	*Co, Stephen* *Foo, James* *Julian, Jedidiah* *Morales, Alejandro Jose*
College: College of Computer Studies	
Department: Software Technology Department	
Course: MS in Computer Science	
Expected duration of project: from: AY Term 3 2025-26 to: AY Term 2 2026-27	

Ethical considerations

The following ethical concerns were identified for this study, together with the steps taken to address them:

1. Vulnerable population (ages 45+). *Careful screening confirms capacity to provide informed consent; consent is explained in plain, accessible language; a researcher is present throughout each session to answer questions or address confusion.*

2. Voluntary participation and informed consent. *Written informed consent is obtained before any data is collected. Participants are informed of the study's purpose, procedures, and their right to withdraw at any time without penalty.*

3. Risk of visual or mental fatigue from eye-tracking and sustained reading tasks. *Each session is limited to approximately 45–60 minutes; a short break is given between the adaptive and static reading conditions; condition order is counterbalanced to offset fatigue effects.*

4. Privacy and confidentiality of gaze and behavioral data. *No personally identifiable information is collected. All recorded data is anonymized/coded and used solely for research purposes, with access restricted to the research team.*

5. Use of the device's front-facing camera for gaze tracking. *The camera is used only to compute gaze coordinates during the reading task; no video/image data is retained beyond what the on-device pipeline needs; capture stops immediately once the task ends.*

6. Audio recording of the post-session debriefing interview. *Recording occurs only with the participant's explicit, separate consent; recordings are transcribed and then handled under the same anonymization and confidentiality protocol as other data.*

7. Data retention. *Data with participant identifiers is retained for no longer than one (1) year after publication of the first paper from the project, after which it is securely destroyed or fully anonymized.*

To the best of our knowledge, the ethical issues listed above have been addressed in the research.

To the best of our knowledge, the ethical issues listed above have been addressed in the research.

<Merlin Suarez>

Name and signature of adviser/mentor

Date:

<name of panelist 1> <name of panelist 2>

Name and signature of panelist Name and signature of panelist Date: Date:

[1]The same form can be used for the reports of completed projects. The appropriate heading need only be used.

DE LA SALLE UNIVERSITY General Research Ethics Checklist
This checklist is to ensure that the research conducted by the faculty members and students of De La Salle University is carried out according to the guiding principles outlined in the Code of Research Ethics of the University. The investigator is advised to refer to the De La Salle University Code of Research Ethics and Guide to Responsible Conduct of Research before completing this checklist. Statements pertinent to ethical issues in research should be addressed below. The checklist will help the researchers and evaluators determine whether procedures should be undertaken during the course of the research to maintain ethical standards. The University's Guide to the Responsible Conduct of Research provides details on these appropriate procedures.

Details of the Research	
Students	Co, Stephen Foo, James Julian, Jedidiah Morales, Alejandro Jose
Thesis Adviser	Merlin Suarez
Department	Department of Software Technology / College of Computer Studies
Title of the Research	A Gaze–Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Older Adults
Term(s) and Academic year in which research is to be conducted	Term 3 AY 2025-2026 to Term 2 AY 2026-2027

This checklist must be completed AFTER the* De La Salle University Code of Ethics *has been read and BEFORE gathering data.

Questions	**Yes**	**No**
1. Does your research involve human participants (this includes new data gathered or using pre-existing data)? If your answer is **yes**, please answer **Checklist A (Human Participants).**	✔	

2. Does your research involve animals (non-human subjects)? If your answer is **yes**, please answer **Checklist B (Animal Subjects).**		✔
3. Does your research involve Wildlife? If your answer is **yes**, please answer **Checklist C (Wildlife)**.		✔
4. Does your research involve microorganisms that are infectious, disease causing or harmful to health? If your answer is **yes**, please answer **Checklist D (Infectious Agents).**		✔
5. Does your research involve toxic/chemicals/ substances/materials? If your answer is **yes**, please answer **Checklist E (Toxic Agents).**		✔

Research with Ethical Issues to address:

If you have a YES answer to any of the above categories, you will be required to complete a detailed checklist for that particular category. A YES answer does not mean the disapproval of your research proposal. By providing you with a more detailed checklist, we ensure that the ethical concerns are identified so these can be addressed in adherence to the University Code of Ethics.

Declaration of Conflict of Interest

[✔] I do not have a conflict of interest in any form (personal, financial, proprietary, or professional) with the sponsor/grant-giving organization, the study, the co-investigators/personnel, or the site.

[ ] I have a personal/family or professional interest in the results of the study (family members who are co proponents or personnel in the study, membership in relevant professional associations/organizations).

Please describe the personal/family or professional interest:

[ ] I have propriety interest vested in this proposal (with the intent to apply for a patent, trademark, copyright, or license)

Please describe propriety interest:

[ ] I have significant financial interest vested in this proposal (remuneration that exceeds P250,000.00 each year or equity interest in the form of stock, stock options or other ownership interests).

Please describe financial interest:

Declaration

We certify that we have read and understand the De La Salle University Code for the Responsible Conduct of Research and will abide by the ethical principles in this document. We will submit a final report of the proposed study to the DLSU-Research Ethics Office. We will not commence with data collection until we receive an ethics review approval from the College Research Ethics Committee.
Name and Signature of Student 1 Name and Signature of Student 2 Name and Signature of Student 3 Name and Signature of Student 4

Endorsement from thesis adviser to the thesis panel for proposal defense…
Name and Signature of Adviser Date

Endorsement from thesis adviser to the thesis panel for final defense… This is to certify that the research was conducted in a manner that adheres to ethical research standards. I am thus endorsing the group for final defense.
Name and Signature of Adviser Date

Guidelines on the Ethical Conduct of Computing Research (v. 2016-09)

De La Salle University	**Research Ethics Review Committee** **Research Ethics Office, 3F Henry Sy Sr. Hall** **De La Salle University Manila** **2401 Taft Avenue, Manila 1004, Philippines** **REO@dlsu.edu.ph (632) 524-4611 loc. 513**	SOP No.: 2 Form No.: 2.03 Version No.: 1 Effectivity Date: July 2016

DE LA SALLE UNIVERSITY
Checklist A
Research Ethics Checklist for Investigations involving Human Participants

This checklist must be completed AFTER the* De La Salle University Code of Research Ethics and Guide to Responsible Conduct of Research *has been read and BEFORE gathering data.* *The University Code of Research Ethics is available at* *http://www.dlsu.edu.ph/offices/urco/forms/URCO Code-of-Research-Ethics_August2011.pdf

NOTE: This checklist is completed after the research proponent fills out the General Checklist Form.

Only answer this Checklist if you answered YES on question 1 of the General Checklist.

Researcher Details	
Students	Co, Stephen Foo, James Julian, Jedidiah Morales, Alejandro Jose
Thesis Adviser	Merlin Suarez
Department	Department of Software Technology / College of Computer Studies
Title of the Research	A Gaze-Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Older Adults

Term(s) and Academic year in which research is to be conducted	Term 3 AY 2025-2026 to Term 2 AY 2026-2027

Provide a brief description of the data collection procedure to be undertaken in the research: Data will be collected in two phases. In a formative study, 3–5 older-adult participants will each complete a one-on-one session with the adaptive reading interface, followed by a brief semi-structured interview; findings will refine the system design and do not contribute to the main research data. In the main study, approximately 10 participants aged 45–65 will be screened (vision check, background/device-usage questionnaire, short reading task), undergo eye-tracker calibration and a baseline-reading recording, then read real news articles under both an adaptive (gaze-driven scaffolding) and a static interface condition in counterbalanced order, answering comprehension questions after each. Gaze data (fixations, saccades, regressions) is captured through the Android device's front-facing camera while a researcher records structured observation notes. Each session concludes with a semi-structured, audio-recorded debriefing interview. All data is anonymized and used solely for research purposes.

De La Salle University	**Research Ethics Review Committee** Research Ethics Office, 3F Henry Sy Sr. Hall De La Salle University Manila 2401 Taft Avenue, Manila 1004, Philippines REO@dlsu.edu.ph (632) 524-4611 loc. 513	SOP No.: 2 Form No.: 2.03 Version No.: 1 Effectivity Date: July 2016

The following should be attached to the checklist:
• A copy of the informed consent form to be used in the study. • A copy of the instrument/tool that will be administered to the participants. • If applicable, a copy of the letter seeking permission to collect data from participants who are under the supervision of an agency, institution, department, or office. • If applicable, a copy of the parental consent form for participants below 18 years old.

The following items refer to important ethical considerations in the conduct of research with human participants. Provide a check for the appropriate answer to each question.

Source of data

Please check all that apply:		
	1. New data will be collected from human participants If you checked this item, how will the new data be gathered? Please check all that apply. **After answering this question, please proceed to page 3**	
		Experimental Procedures/Intervention/ Treatments
		Focus Group
	✔	**Personal Interviews**
		Self-administered Questionnaire
	✔	**Researcher-administered Questionnaire**
		Internet survey
	✔	**Observation**
		Telephone survey
	✔	**Others, please specify: Eye-tracking / gaze-behavior recording (fixations, saccades, regressions) during reading tasks**
	2. Pre-existing data from human participants, i.e., from a dataset If you checked this item, please proceed to page 7	

If both options are checked (both new data and pre-existing data), **answer all of the questions in this document.**

ONLY ANSWER IF NEW DATA WILL BE COLLECTED (item 1 above)

Sampling Details	
Number of Participants/Subjects	**15**
Location where the participants will be recruited/ where subjects will be obtained?	**Online, Through Family and Friends**
How long will the data collection take place?	**4 months**

Who will perform the data collection?	**Students/Researchers**
Location(s) where data collection will take place	**Online and In-person Meetings**
What procedures will be employed to ensure voluntary consent from participants?	**Participants are briefed on the study's purpose and procedures and informed of their right to withdraw at any time without penalty; written informed consent is obtained before any data is collected; no coercion or pressure is applied, and declining or withdrawing carries no penalty.**
Data Retention	
How long will data with participant identifiers be kept after the publication of the first paper from the project?	**One (1) Year**
How long will anonymized data be kept after the publication of the first paper from the project?	**One (1) Year**
Procedure for Informed Consent	
How will informed consent be recorded? (check all that applies) Reminder: please attach informed consent that will be used in the study	[✔] Written Consent [ ] Audio-recorded Consent [ ] Online/Email recorded Consent [ ] Others, please specify:

If you will not obtain a recorded informed consent, answer the questions that follow:

Why does the waiver of informed consent not pose a threat to the welfare and rights of the participants?

Why is recording an informed consent not practical for the proposed study?			
	Yes	**No**	**Not Applicable**
1. Will the research involve students who will be receiving course credits for their participation? **If YES, please attach a copy of the consent form and a summary of the debriefing process that will help participants understand how their participation in the research has provided a relevant learning experience to the crediting course.**		✔	
2. Does the study involve participants below 18 years old or those who are unable to give their informed consent? **If YES, please attach a copy of the parental consent form.**		✔	

3. Is there a possibility that the research can induce physical and/or psychological harm to the participants? Will they experience pain or some discomfort as a result from their participation in the research? **If YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.**	✔		Benefit (improving digital reading accessibility for older adults) outweighs this minor, fully reversible risk, which is mitigated by limiting sessions to 30 minutes, providing a break between conditions, and allowing withdrawal at any time.
4. Will the participants be deliberately falsely informed or made unaware that they are being observed? Will they be misled in a way that they will possibly object to or show unease when told of the real purpose of the study? **If YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.**		✔	
5. Will the research involve the discussion of, or questions on, sensitive topics (e.g. sexual activity, substance abuse, or mental health)? **If YES, please make sure that the informed consent form explicitly states that sensitive questions will be posed and that you will safeguard the anonymity of the participants and ensure confidentiality. Please attach a copy of your informed consent form and your instrument.**		✔	
6. Will the research involve the administration of drugs, or other substances to the participants?		✔	

If YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants. **Please also attach a description of the procedure that will ensure that the participants will be brought back to their physical and psychological states prior to their participation in the research.**			
7. Will biological samples (e.g. blood, saliva, urine) be obtained from the participants? **If YES, will this involve invasive procedures? Please attach a description of these procedures.**		✔	
8. Will genetic materials be obtained from the biological samples? **If YES, please attach a description of the procedures that will ensure confidentiality. Please attach the informed consent form.**		✔	
9. Will financial inducements (other than reasonable expenses, like transportation or meal allowances) be offered to the participants for their participation in their research? **If YES, the researcher(s) should be mindful of how the inducements can influence the participants' responses or behaviors during the research. Indicate the financial inducements offered to the participants:** ______________________________________		✔	
10. Is there a possibility for groups or communities to be harmed by the dissemination of the research findings? **If YES, please attach a description of procedures to ensure the anonymity and confidentiality of the research findings.**		✔	

5

Answering YES to most of the above items will signal an ethical issue that needs to be addressed. Some actions that will allow adherence to research ethical principles are provided with each item. The researcher is advised to refer to the University's Guide to the Responsible Conduct of Research for the appropriate procedures to ensure adherence to ethical principles in the conduct of research.

Declaration

We certify that we have read and understand the De La Salle University Code for the Responsible Conduct of Research and will abide by the ethical principles in this document. We will submit a final report of the proposed study to the DLSU-Research Ethics Office. We will not commence with data collection until we receive an ethics review approval from the College Research Ethics Committee.

Name and Signature of Student 1

Name and Signature of Student 2

Name and Signature of Student 3

Name and Signature of Student 4

Endorsement from thesis adviser to the thesis panel for proposal defense…

Name and Signature of Adviser Date

Endorsement from thesis adviser to the thesis panel for final defense…

This is to certify that the research was conducted in a manner that adheres to ethical research standards. I am thus endorsing the group for final defense.

Name and Signature of Adviser Date

6

DE LA SALLE UNIVERSITY

College of Computer Studies – Department of Software Technology

INFORMED CONSENT FORM

A Gaze–Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Middle-Aged and Older Adults

Researchers: Co, Stephen; Foo, James; Julian, Jedidiah; Morales, Alejandro Jose

Thesis Adviser: Merlin Suarez

Department: Software Technology Department, College of Computer Studies, De La Salle University

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1. Purpose of the Study

You are being invited to take part in a research study about how adults aged 45 and above read news on a smartphone, and how the design of a reading application affects that experience.

The study looks at a reading application that adjusts its display while you read. You will read using two versions of it: one that changes its appearance as you read, and one that keeps the same appearance throughout. We are interested in how the reading experience differs between the two.

This form explains what will happen if you agree to take part, so that you can decide whether or not you want to participate. At the end of the session, once all the reading and questions are finished, we will explain the study to you in fuller detail.

2. What You Will Be Asked to Do

If you agree to participate, you will be asked to:

- Complete a short vision check and a brief questionnaire about your background and familiarity with digital devices.
- Undergo a short calibration process using your device's front-facing camera so the system can track your eye movements accurately.
- Read a set of short news articles on an Android smartphone or tablet, once with the adaptive (gaze-responsive) interface and once with a standard, non-adaptive interface, in an order determined by the researchers.
- Answer a few simple comprehension questions after each reading passage.
- Take part in a short interview afterward about your experience with the reading interface.

The entire session, including breaks, is expected to last approximately 45–60 minutes. A short break will be provided between the two reading conditions.

3. Use of the Device Camera

The study uses your device's front-facing camera to estimate where you are looking on the screen while you read. The camera is used only to compute gaze coordinates during the reading task. No video or image recordings of your face are stored beyond what the on-device processing pipeline requires, and camera capture stops immediately once the reading task ends.

4. Audio Recording

With your separate, explicit permission, the post-session interview will be audio-recorded and later transcribed. You may decline to have the interview recorded and still participate in the rest of the study. Recordings are handled under the same confidentiality protections described below.

5. Voluntary Participation and Right to Withdraw

Your participation in this study is completely voluntary. You may choose not to participate, or you may withdraw from the study at any point, for any reason, without any penalty or negative consequence. If you choose to withdraw, any data collected from you up to that point will be discarded upon request.

6. Possible Risks and Discomforts

The study involves minimal risk. You may experience mild visual fatigue or eye strain from the reading and calibration tasks. To reduce this risk, session length is limited to approximately 45–60 minutes, a break is provided between reading conditions, and you may pause or stop at any time. Any discomfort experienced is expected to be minor and fully reversible.

7. Benefits

You may not directly benefit from taking part in this study. However, your participation will contribute to research aimed at improving the accessibility of digital reading tools for adults aged 45 and above, which may benefit you and others in the future.

8. Privacy and Confidentiality

No personally identifiable information will be collected as part of the study data. All recorded data, including gaze data, questionnaire responses, and interview transcripts, will be anonymized or coded and used solely for research purposes. Access to your data will be restricted to the research team.

Data that includes participant identifiers (such as signed consent forms) will be retained for no longer than one (1) year after the publication of the first paper from this project, after which it will be securely destroyed. Anonymized research data will likewise be retained for no longer than one (1) year after publication before secure disposal.

9. Who to Contact

If you have any questions about this study, or if you experience any concerns related to your participation, you may contact the research team.

Co, Stephen – tikoy_co@dlsu.edu.ph

Foo, James – james_foo@dlsu.edu.ph

Julian, Jedidiah – jedidiah_kyle_julian@dlsu.edu.ph

Morales, Alejandro Jose – alejandro_morales@dlsu.edu.ph

10. Statement of Consent

Please read the statements below and indicate your agreement by checking the boxes and signing where indicated.

☐ I have read (or had read to me) the information above, and I understand the purpose and procedures of this study.

☐ I understand that my participation is voluntary and that I may withdraw at any time without penalty.

☐ I understand that the device's front-facing camera will be used to track my eye movements during the reading tasks.

☐ I understand that some details of the study will be explained more fully at the end of the session, and that I may decide at that point whether my data is used.

☐ I understand that no personally identifiable information will be collected and that my data will be anonymized.

☐ I agree to have the post-session interview audio-recorded. (optional — you may still participate if you leave this unchecked)

☐ I voluntarily agree to take part in this study.

____

Participant's Printed Name

____

Participant's Signature Date

____

Researcher's Signature Date

A copy of this signed consent form will be provided to you for your records.

The debriefing script and screening guide that follow were added in the August 2026 revision.

DE LA SALLE UNIVERSITY
College of Computer Studies, Department of Software Technology

PARTICIPANT DEBRIEFING SCRIPT

A Gaze–Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Middle-Aged and Older Adults

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Delivery rules, read before first use

Deliver only after the usability survey and the interview are both finished. Debriefing before the interview contaminates the qualitative data: once someone has been told the display responded to signs of losing their place, every answer afterwards is an answer about what they were just told. The session order is:

reading conditions → **usability survey** → **interview** → **debriefing**

Say the quoted lines close to as written. Four researchers deliver this script and it needs to land the same way every time. Sound natural, do not read it woodenly, but do not change the substance, especially in Sections 3 and 4. Bracketed text is a note to you, not something to read out.

Sessions are in English. The Filipino line under each English one is there if the participant is more comfortable that way, or if they look like they are not following. Use whichever fits the person in front of you, or mix them. If you switch to Filipino, still do not paraphrase Sections 3 and 4, and note on the form that you did.

Do not soften it, and do not oversell it. Some people will feel a bit silly for not noticing. Section 3 handles that. Do not go further and imply their data was disappointing or that they did anything wrong.

Section 4 is required and gets recorded for everyone, including people who are obviously happy to continue.

Everyone gets debriefed, including anyone who stopped partway and anyone excluded after enrolment. Someone who left early still consented without the full picture, so they are still owed it.

---

1. What we measured

[Reading done, survey and interview finished. Sit at their level. Phone face down or app closed, no screen between you.]

"Thanks, that's everything I needed you to do. Before we wrap up, I want to explain the study a bit more than I did at the start. It takes about five minutes, and then I have one question to ask you about your data."

"Salamat po, tapos na po tayo sa lahat. Bago tayo magtapos, gusto ko lang pong ipaliwanag nang kaunti pa ang pag-aaral kaysa sa sinabi ko kanina. Mga limang

minuto lang po, tapos may isa akong itatanong tungkol sa data ninyo."

"At the start I told you we were looking at how a reading app that adjusts its display affects the reading experience. That was true, but it wasn't the whole story, and I want to fill it in now."

"Kanina po sinabi ko na tinitingnan namin kung paano nakakaapekto sa pagbabasa ang isang app na nagbabago ang display. Totoo po iyon, pero hindi po iyon ang buong kuwento, at gusto ko na pong kumpletuhin ngayon."

"Here's what we were really measuring. While you're reading, something interrupts you. You look away, or you lose your spot on the page. What we wanted to know is how long it takes you to settle back into reading afterwards. How quickly you pick up where you left off. That was the main thing this study was after, and we measured it from your eye movements."

"Ito po ang talagang sinusukat namin. Habang nagbabasa kayo, may pumuputol sa atensyon ninyo. Baka tumingin kayo sa ibang direksyon, o mawala kayo sa pinagbabasahan ninyo. Ang gusto naming malaman ay kung gaano katagal bago kayo makabalik sa dati ninyong pagbabasa. Gaano kabilis kayong makabalik sa huli ninyong binabasa. Iyon po ang pangunahing hinahanap ng pag-aaral na ito, at sinukat namin iyon sa galaw ng mga mata ninyo."

2. What the app was doing

"The version of the app that changed how it looked, the highlighting and the dimming, wasn't doing that randomly. It wasn't on a timer either."

"Iyong bersyon ng app na nagbabago ang itsura, iyong pag-highlight at pagdilim, hindi po iyon basta-basta nagbabago. Hindi rin po iyon naka-timer."

"It was following your eye movements while you read. When the pattern looked like you might have lost your place, that's when it changed the display, to help you find your way back. The other version never did that. It stayed the same the whole time, no matter what."

"Sinusundan po niya ang galaw ng mga mata ninyo habang nagbabasa kayo. Kapag mukhang nawala kayo sa pinagbabasahan ninyo, doon po nagbabago ang display, para matulungan kayong makabalik. Iyong isang bersyon naman, hindi po gumagawa niyon kailanman. Pareho lang po iyon buong oras, kahit ano pa ang mangyari."

[If they ask what it was looking for specifically: the app compares how they were reading recently against how they were reading earlier in the session, and reacts when the two differ. Do not go further into the algorithm unless they ask again.]

3. Why we didn't say so at the start

"The reason I didn't explain that beforehand is that knowing it changes what people do."

"Ang dahilan po kung bakit hindi ko iyon ipinaliwanag kanina ay dahil kapag alam ninyo, nagbabago ang ginagawa ng tao."

"If you'd known the app was watching for the moments you lost your place, it's pretty natural to start checking whether you're losing your place. Or to wait for the highlight instead of just reading. And that's exactly the thing we were trying to measure. So we'd have ended up measuring people watching themselves read, instead of people reading."

"Kung alam ninyo pong binabantayan ng app kung kailan kayo nawawala sa binabasa, natural lang pong simulan ninyong i-check kung nawawala nga kayo. O maghintay sa highlight imbes na basta magbasa lang. At iyon mismo po ang sinusubukan naming sukatin. Kaya ang masusukat lang namin ay mga taong binabantayan ang sarili nilang pagbabasa, hindi mga taong nagbabasa talaga."

"So we described the study accurately, just not completely. I want to be clear about what that did and didn't mean. Everything I told you at the start was true. The camera, the eye tracking, the two versions, how long it would take, what happens to your data. None of it was false, and nothing was hidden from you about what we were recording. What we held back was which measurement mattered most to us, and what made the display change."

"Kaya tama po lahat ng sinabi namin tungkol sa pag-aaral, hindi nga lang kumpleto. Gusto ko pong linawin kung ano ang ibig sabihin niyon at kung ano ang hindi. Totoo po lahat ng sinabi ko kanina. Ang camera, ang pagsubaybay sa mata, ang dalawang bersyon, kung gaano katagal, kung ano ang mangyayari sa data ninyo. Wala po akong sinabing hindi totoo, at walang itinago tungkol sa kung ano ang nire-record namin. Ang hindi lang po namin sinabi ay kung aling sukat ang pinakamahalaga sa amin, at kung ano ang dahilan ng pagbabago ng display."

"And just so you know, there was no right or wrong way to read today. Not noticing the changes isn't a mistake. For a lot of people, that's the point."

"At para po malaman ninyo, walang tama o maling paraan ng pagbabasa kanina. Hindi po pagkakamali kung hindi ninyo napansin ang mga pagbabago. Para po sa marami, iyon nga ang punto."

4. Your data, required, record the answer

"Now that you know all that, I need to ask you again about your data."

"Ngayong alam na po ninyo iyon lahat, kailangan ko lang pong itanong ulit tungkol sa data ninyo."

"You can say no. If you do, I'll delete everything from your session today. The eye tracking data, your answers, and the interview recording. There's no penalty, nothing changes about you being here, and you don't have to give me a reason."

"Puwede po kayong tumanggi. Kung tatanggi kayo, buburahin ko po ang lahat ng galing sa session ninyo ngayon. Ang data ng eye tracking, ang mga sagot ninyo, at ang recording ng panayam. Wala pong parusa, walang magbabago sa pagpunta ninyo dito, at hindi po ninyo kailangang magbigay ng dahilan."

"So, knowing now what the study was really measuring, is it still okay with you if we use your data?"

"Kaya po, ngayong alam na ninyo kung ano talaga ang sinusukat ng pag-aaral, okay pa rin po ba sa inyo na gamitin namin ang data ninyo?"

[**If NO:** thank them properly, do not try to talk them round, do not ask why. Delete the session data in front of them if they want to see it. Log the exclusion as "withdrawn post-debriefing". They get any compensation or thanks exactly as they otherwise would.]

[**If they hesitate**, or ask what happens either way: repeat the two options plainly and wait. Do not fill the silence. A hesitant yes given under social pressure is not consent.]

DEBRIEFING RECORD, completed for every participant	
Participant ID	________________
Consent for data use maintained after debriefing	☐ Yes ☐ No
Delivered in	☐ English ☐ Filipino ☐ Both
Participant initials	________________
Researcher initials	________________
Date and time	________________

5. Questions and contacts

"Any questions? About the study, the app, or what happens to your data now?"

"May tanong po ba kayo? Tungkol sa pag-aaral, sa app, o sa kung ano ang mangyayari sa data ninyo ngayon?"

[Answer plainly and fully. There is nothing left to hold back at this point. If you do not know, say so and offer to follow up rather than guessing.]

"If anything comes to mind later, you can reach any of us here."

"Kung may maisip po kayo mamaya o bukas, puwede ninyo pong kontakin kahit sino sa amin dito."

[Hand over the contact slip.]

Co, Stephen, tikoy_co@dlsu.edu.ph
Foo, James, james_foo@dlsu.edu.ph
Julian, Jedidiah, jedidiah_kyle_julian@dlsu.edu.ph
Morales, Alejandro Jose, alejandro_morales@dlsu.edu.ph

"Thank you, really. This is the part of the research we can't do without."

"Maraming salamat po talaga. Ito po ang bahagi ng pananaliksik na hindi namin kayang gawin nang wala kayo."

If they say any of these

They say	You say
"So you were testing me?"	"No, we were testing the app. There was nothing you could get right or wrong." *"Hindi po, ang app po ang sinusubukan namin. Walang po kayong puwedeng maling masagot."*
"I didn't notice any changes."	"That's completely normal, and it's useful to us either way." *"Normal na normal po iyon, at nakakatulong pa rin po sa amin."*
"Why not just tell people?"	Repeat Section 3 briefly. Do not apologise for the design. It was reviewed and approved.
"Did I do badly?"	"There's no scoring of you at all here. We're measuring the app." *"Wala po kaming ini-score sa inyo. Ang app po ang sinusukat namin."*
"Can I change my mind later?"	"Yes, email any of us and we'll delete your data." This is true, so honour it. *"Opo, mag-email lang po kayo sa kahit sino sa amin at buburahin namin ang data ninyo."*
They seem annoyed at the omission	Do not get defensive. "That's fair, and it's exactly why I'm asking you now instead of just assuming." Then go to Section 4 and take whatever answer they give. *"Tama po kayo, at iyon nga po ang dahilan kung bakit tinatanong ko kayo ngayon imbes na basta na lang ipagpalagay."*

DE LA SALLE UNIVERSITY

College of Computer Studies, Department of Software Technology

PARTICIPANT SCREENING GUIDE

A Gaze–Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Middle-Aged and Older Adults

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How to use this guide

Work through Sections 1 to 8 in order. **Stop at the first failed criterion** and record it. Every exclusion gets a reason written down (Section 9).

Sessions are in English. The Filipino line under each script is there if the person is more comfortable that way. Bracketed text is a note to you, not read aloud.

On how you talk about age. Participants were not recruited for being old. They were recruited because the study needs readers across a range of ages and the range starts at 45. Say "we're recruiting from 45 up" rather than anything that frames them as elderly, and when explaining the vision check, say we check *everyone*. Nobody should leave feeling they were picked out as a case study in ageing. The academic title above uses "middle-aged and older adults" and that is fine on paper; do not use it out loud.

Do not elaborate on Section 1 of the consent form. If asked what the display changes do, use the prepared line in Section 3. Improvising here breaks the study.

1. Eligibility and opening

> "Thanks for coming in. Before we start I need to run through a few quick questions to check the study's a good fit. Takes about ten minutes. Some of it's about your eyes and your reading, because the whole study is reading on a phone. You can skip anything, or stop, whenever you like."
>
> *"Salamat po sa pagpunta. Bago tayo magsimula, may ilan lang po akong itatanong para makita kung bagay sa inyo ang pag-aaral. Mga sampung minuto lang po. May ilan pong tungkol sa mata ninyo at sa pagbabasa, kasi pagbabasa po sa cellphone ang buong pag-aaral. Puwede po kayong lumaktaw o huminto kahit kailan."*
>
> "Can I confirm you're 45 or over? We're recruiting from 45 up for this one."
>
> *"Makukumpirma ko lang po, 45 pataas na po kayo? Mula 45 pataas po kasi ang hinahanap namin dito."*

Record the age band, not the birth date. **There is no upper limit.** Do not turn anyone away for being older; Sections 4, 5 and 8 handle suitability directly.

2. Consent

Read the Informed Consent Form aloud in plain language. Then, before they sign:

"Just so I know I've explained it properly, can you tell me in your own words what you'll be doing today, and what happens if you decide partway through that you'd rather stop?"

"Para lang po masiguro kong maayos kong naipaliwanag, puwede ninyo pong sabihin sa sarili ninyong salita kung ano ang gagawin natin ngayon, at kung ano ang mangyayari kung gusto ninyong huminto sa gitna?"

This is the capacity check. They should convey, in any words: reading on a phone while the camera follows their eyes, questions afterwards, and that they can stop anytime without consequence.

- Gets both → proceed.
- Unclear → explain again differently, ask once more.
- Still unable, or cannot state the right to stop → do not enrol. Record "capacity check not met". Thank them warmly and do not explain the exclusion in terms that suggest a deficit.

Signature only after this. Give them their copy.

[If they ask what the display changes do, or when: "That's part of what we'll go through at the end. I'd rather not shape how you read by explaining it now. Is that all right?" Do not add anything to this.]

3. Near vision check

You do not need any eye-care training to do this. It is a screening check, not an eye examination. You are not diagnosing anything. You are recording how small a line of print the person can read, so we can rule out that poor vision, rather than reading behaviour, explains what we measure.

The terms, in plain language

Near acuity	How small a print size someone can read *up close*, at about arm's length or nearer. Different from distance vision. Someone can drive fine and still struggle with small print.
Correction	Any glasses or contact lenses. "Corrected" means wearing them, "uncorrected" means not.
Habitual correction	Whatever that person would *normally* wear for this task. Not their newest pair, not their best pair. What they would actually have on if they picked up their phone at home.
Reading glasses	Glasses worn only for close work. Many people wear them for books or newspapers but not for a phone, which they hold closer. This matters to us, so ask specifically.
Progressives / bifocals	One lens with different strengths in different zones. The wearer tilts their head to find the reading zone near the bottom.
20/40	A way of writing acuity. The smaller the second number, the sharper the vision. 20/20 is the usual reference point; 20/40 is somewhat softer; 20/200 is considerably worse. We accept 20/40 or better.

The card

The Rosenbaum pocket card is a small card with rows of numbers that get smaller down the card. Along the side of each row are two sets of labels: a **J number** (J1, J2, J3 and so on) and a **20/** number (20/20, 20/25, 20/30 and so on). Both label the same row. **We use the 20/ numbers and ignore the J numbers.**

Before the first session, twice-check the card:

1. If it was printed rather than bought, make sure it was printed at 100 percent with no “fit to page” scaling. A card printed even slightly small makes everyone look better than they are, and every reading you take with it is wrong. Most cards have a reference scale printed on them; measure it with a ruler.
2. Cut a piece of string or ribbon to the distance printed on the card, usually 14 inches or 35 centimetres. Tape one end to the card. This is your distance cord and it is used every time.

If they wear glasses

The rule: they wear whatever they would normally wear to read on their phone, and they keep it on for the whole session. Ask the question that way, not “do you wear glasses”. Many people have a pair for print that they do not bother with for a phone, because they hold the phone closer.

Situation	**What to do**
Reads phone without glasses	Test without. Record “none”. Fine even if they use readers for books.
Reads phone with reading glasses	Test with them on. Record “reading only”.
Wears progressives or bifocals all day	Test with them on. Record “progressive”. See the note below.
Contact lenses	Test with them in. Record “contacts”.
Two pairs and they are unsure	Ask which one they would pick up if they were reading the news on their phone at home. Use that. Do not let them switch later.
Left their glasses at home	**Do not test uncorrected and carry on.** If they normally need them for phone reading, the acuity reading and the calibration will both describe someone who is not them. Reschedule, or run the session and mark it excluded.

Whatever they wear, it stays on from the vision check through to the end of the reading task. This matters more here than in an ordinary vision screening. The eye tracker is calibrated to how their eyes look to the camera, so swapping or removing glasses partway invalidates the calibration and every measurement after it. Check again before you start the reading task, in Section 9.

> [**Progressives and bifocals:** the wearer tilts their head to find the reading zone near the bottom of the lens. That head position must be the same during calibration and during reading, so let them settle into it *before* you calibrate, not after. Watch for the drift flag at the calibration gate,

which is what catches it if they shift.]

[**Glasses and the tracker:** lens reflections can interfere with the camera finding the eyes, and this has not been tested on the target age group. You do not need to judge this yourself. The calibration gate in Section 8 will catch it, and a red band is the signal, not anything you can see. Record that glasses were worn so the pattern can be checked later across participants.]

Setting up

"We check everyone's near vision before we start, since the whole thing is reading on a phone. Wear whatever you'd normally wear to read on your phone. This isn't an eye test, I'm just checking the print size works for you."

"Sinusuri po namin ang malapitang paningin ng lahat bago magsimula, kasi pag-babasa po sa cellphone ang buong pag-aaral. Isuot lang po ninyo kung ano ang normal ninyong suot kapag nagbabasa sa cellphone. Hindi po ito eye test, tinitingnan ko lang po kung kaya ninyong basahin ang laki ng letra."

- Seat them comfortably, in good light. Light should fall **on the card**, not into their eyes and not from behind them.
- Confirm they are wearing what they would normally wear to read a phone. If they take glasses out of a bag, ask whether those are the ones they use for their phone specifically.
- Hand them the card. Have them hold the free end of the cord lightly against their cheekbone, just under one eye, so the card sits at the correct distance. Check the cord is straight and not slack.

Running the check

Test in this order: **right eye, then left eye, then both together.**

1. **Right eye.** Ask them to cover the left eye with their palm, cupped, not pressed. Pressing on an eye blurs it briefly and spoils the next reading.
2. Ask them to read the smallest row they can manage, out loud, left to right.
3. Move up a row if they cannot, down a row if they get it easily. Stop at the smallest row they can still read.
4. **A row counts as read if they get more than half the characters right.** Apply this the same way every time; do not let "close enough" drift between researchers.
5. Write down the 20/ label of that smallest row.
6. Repeat for the left eye, palm over the right.
7. Repeat with **both eyes open**. This is the reading used for the pass or fail decision.

"Cover your left eye with your palm, don't press. Now read me the smallest line you can, going left to right. Guessing is fine."

"Takpan po ninyo ang kaliwang mata gamit ang palad, huwag pong idiin. Basahin po ninyo ang pinakamaliit na linyang kaya ninyo, mula kaliwa pakanan. Okay lang pong manghula."

Reading the result

Both eyes 20/40 or better	Pass. Record and continue.
Both eyes worse than 20/40	Do not enrol. Record the reading and "near acuity below threshold".
Cannot read even the largest row	Do not enrol. Record it plainly and move on. Do not comment further.

20/40 is *better* than 20/50. If the numbers confuse you, the rule is simply: the second number must be 40 or lower.

Things that go wrong

- **Squinting.** Squinting sharpens vision temporarily and inflates the result. Ask them to relax their eyes and read again.
- **Memorising.** If they read the same rows repeatedly, they may recall the numbers. Most cards have alternative rows; use a different one for the second eye if available.
- **The cord going slack.** Check it each time. This is the most common error.
- **Progressive or bifocal wearers** need a moment to find the reading zone, usually by tilting the head back slightly. Let them settle. Do not take the first hesitant line as their limit.
- **They ask what their result means.** "It just tells me the print size works for you, that's all I'm checking." Do not interpret it, do not compare it to normal, and do not suggest they see anyone about it.

> [Whatever the result, the same correction they wore for this check must be worn for the reading task later. Note it and check again before Section 8.]

4. Health exclusions

> "Has an eye doctor ever diagnosed you with any of these?"
>
> *"Mayroon po bang na-diagnose sa inyo na doktor sa mata na ganito?"*

Exclude on any **diagnosed** condition below. You do not need to understand these clinically; you need to recognise the name if they say it, and you need to know that any of them is an exclusion.

Cataract	Clouding of the lens, common with age. Excluded only if untreated or still affecting vision. Many people have had it removed and see well afterwards; that is not an exclusion, so ask whether it was treated.
Macular degeneration	Loss of central vision, the part used for reading. Excluded.
Glaucoma	Pressure-related damage. Excluded if they mention any loss of side or field vision. Glaucoma managed with drops and no reported loss is a judgement call; ask whether a doctor has told them they have lost any vision.
Diabetic retinopathy	Damage to the back of the eye from diabetes. Excluded. Note that diabetes by itself is *not* an exclusion.
Anything else diagnosed	Any other condition a doctor has told them affects their reading vision. Excluded.

> "And has a doctor diagnosed anything neurological that affects your reading, concentration or memory?"
>
> *"May na-diagnose po bang doktor na kondisyon sa utak o nerbiyos na nakakaapekto sa pagbabasa, konsentrasyon o memorya ninyo?"*

Exclude on any diagnosed condition affecting reading, concentration or memory. Examples they might name: stroke affecting vision or reading, dementia or Alzheimer's, Parkinson's disease, multiple sclerosis, a significant head injury with lasting effects, or a diagnosed reading disorder such as dyslexia.

Ask about diagnoses, not symptoms. You are not assessing anyone's health, and you are not qualified to. The question is always "has a doctor told you", never "do you have trouble with". If they volunteer a symptom with no diagnosis, write it in the notes and enrol if Section 3 passed.

> [If excluded here, do not speculate about their health or tell them to see a doctor. "This one needs a pretty narrow set of conditions and it isn't a fit, but thank you, genuinely."]

5. Background questionnaire

Researcher-administered. Fields: age band; education completed; reading language and preference for today; smartphone use frequency; how often they read news on a phone; typical reading stretch in one sitting; whether they adjust font size on their own phone and to what; any prior experience with eye tracking.

6. Short reading task

A held-out NewsMead article, roughly 150 to 200 words, shown in the app at the study font size, read silently, followed by one comprehension question.

Not a reading test. It confirms the passages are readable for this person and gives them a low-stakes run-through before anything is recorded. The article used is named in the study record and is excluded from the reading conditions for everyone.

7. Calibration gate

Run the 16-point calibration from the home screen. The app reports leave-one-out and validation error in line heights and bands the result.

Band	Median error	Do this
Green	At or below 3.0 line heights	Accept, proceed
Amber	At or below 4.5 line heights	Accept, proceed
Red	Above 4.5 line heights	Redo all once, after a short rest

- Red twice on full attempts → do not proceed. Record "calibration gate not met".
- Drift flag raised → redo whatever the band. They moved during calibration.
- "Redo worst points" can be used freely within an attempt. It does not count as a fresh attempt.

Record band, median error in line heights, and number of full attempts for everyone. **Do not accept a red calibration at your discretion during the main study.** The app allows it and logs it; that is for development, not for participant data.

Anyone excluded here is still debriefed and still receives any compensation.

8. Session briefing

Given after enrolment, immediately before the reading task.

> "Okay, you're all set. Here's what happens next. You'll read a few short news articles on this phone, with a couple of quick questions after each one. There are two versions of the app and you'll use both, with a short break in between. All in, about forty-five minutes to an hour."
>
> *"Ayos po, handa na tayo. Ganito po ang susunod. Magbabasa po kayo ng ilang maiikling balita dito sa cellphone, may ilang mabilis na tanong pagkatapos ng bawat isa. May dalawang bersyon po ang app at gagamitin ninyo pareho, may maikling pahinga sa gitna. Lahat-lahat po, mga apatnapu't limang minuto hanggang isang oras."*
>
> "Hold the phone however's comfortable, about this far from your face. Once we start, try to keep it roughly there and stay more or less settled, because the camera's following your eyes and it works best if things don't shift around too much. You don't have to freeze. Just get comfortable now rather than halfway through."
>
> *"Hawakan po ninyo ang cellphone kung saan kayo komportable, mga ganito po kalayo sa mukha ninyo. Kapag nagsimula na, subukan po ninyong panatilihin iyon at huwag masyadong gumalaw, kasi sinusundan ng camera ang mga mata ninyo at mas maganda kung hindi masyadong nagbabago ang posisyon. Hindi naman po kailangang manigas. Ayusin lang po ninyo ngayon ang pagkakaupo kaysa sa gitna na."*

[Show the distance with your hand rather than stating a number. Check their seating and the lighting before you start.]

"If you need to stop, stretch, or take a break, just say so. Any time, not just at the break. And while you're reading I'll be here but I won't be helping, so don't worry that I'm being unfriendly. It's just so everyone's reading the same way."

"Kung kailangan ninyong huminto, mag-unat, o magpahinga, sabihin lang po ninyo. Kahit kailan po, hindi lang sa break. At habang nagbabasa kayo, nandito lang po ako pero hindi ako tutulong, kaya huwag po ninyong isipin na masungit ako. Para lang po pareho ang takbo para sa lahat."

"Just read the way you normally would. Ready when you are."

"Magbasa lang po kayo gaya ng nakasanayan ninyo. Kayo na po ang bahala kung kailan tayo magsisimula."

Researcher checklist before starting

- Same correction worn as during the vision check
- Phone at roughly the calibration distance, participant settled
- Lighting adequate and not behind the participant
- `GAZE_TOUCH_VALIDATION` false, `SCAFFOLD_MODE` adaptive
- Debug visuals off for participant-facing runs
- First 20 seconds are baseline collection; no scaffold appears

9. Enrolment record

Completed for everyone screened, enrolled or not.

Screening ID	______________________
Date / Researcher	______________________
Aged 45 or above (Section 1)	☐ Yes ☐ No
Consent capacity confirmed (Section 2)	☐ Yes ☐ No

Near vision check (Section 3)

Smallest row read	**Write 20/___**	
Right eye	20/____	
Left eye	20/____	
Both eyes together	**20/____**	**this one decides pass or fail**

Both eyes 20/40 or better	☐ **PASS** ☐ **FAIL**
Was correction worn during the check?	☐ No ☐ Glasses ☐ Contacts
If glasses, what kind?	☐ Reading only ☐ Distance ☐ Progressive or bifocal
Does the participant wear these when reading on their *phone* specifically?	☐ Yes ☐ No ☐ Sometimes
Same correction confirmed worn for the reading task (check again at Section 8)	☐ Yes
Health exclusions (Sections 4 and 5)	
Any diagnosed eye condition affecting reading	☐ None reported ☐ Yes, exclude
Any diagnosed neurological condition	☐ None reported ☐ Yes, exclude
If either yes, which	______________________
Symptoms mentioned without a diagnosis	______________________
Remaining steps and decision	
Questionnaire and reading task done (6, 7)	☐ Yes
Calibration band (Section 8)	☐ Green ☐ Amber ☐ Red
Median error in line heights / full attempts	______________________
Session briefing delivered (Section 9)	☐ Yes
Decision	☐ Enrolled ☐ Not enrolled
Reason if not enrolled	______________________

Screening records carry identifiers. Retain for one year after first publication, then destroy securely, same as the signed consent forms.

Appendix B

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StabiliRead: A Gaze–Driven Adaptive Reading Interface to Reduce Recovery Cost During Sustained Digital Reading in Middle-Aged and Older Adults

15 A Thesis Proposal

presented to

the Department of Software Technology
College of Computer Studies
De La Salle University

In partial fulfillment
of the requirements for the degree of

Bachelor of Science in Computer Science
Major in Software Technology

by

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Abstract

Digital reading is now one of the main ways people access information. Unlike traditional reading, digital environments are dynamic. Text is often surrounded by other on-screen elements, such as notifications and pop-ups, which can interrupt how we read. Readers, particularly middle-aged and older adults, often experience difficulty navigating this environment due to their increased susceptibility to distraction and slower processing speeds, resulting in increased cognitive effort and reduced reading efficiency. This study aims to address the lack of adaptive graded-based systems that support recovery during disrupted reading. To address this gap, the study proposes a personalized gaze-driven adaptive reading interface that provides graded visual scaffolding based on moment-to-moment reading behavior. The system is implemented as an extension of NewsMead, an existing Android news-reading application designed for older Filipino adults. This system introduces a stability index derived from gaze metrics to detect instability and dynamically adjust visual support in order to assist re-orientation and maintain reading continuity. The study will be conducted using eye-tracking to capture gaze behavior and identify instability indicators, and evaluate system effectiveness through measures such as fixation duration, saccades, and regression, indicating stability and recovery latency. By focusing on middle-aged and older adults (aged 45 and above), where reading instability is often more pronounced the study aims to provide a clearer evaluation of recovery-focused support and contribute to a more responsive and human-centric digital reading systems.

Keywords: digital reading, gaze-based interaction, visual scaffolding, reading stability, adaptive interfaces

Contents

Appendix C

Gaze-Estimation Measurements

This appendix provides the supporting gaze-estimation measurements used in Chapter 4.

Supporting data for the method comparison in Section 4.3.2.

C.1 Conditions

Samsung Galaxy A56 (SM-A566B), 1080 by 2340 pixels over 7.10 by 15.30 centimetres, approximately 152 pixels per centimetre, front camera 3.55 cm from the left edge and 0.38 cm from the top of the display. Held in portrait orientation. Device geometry measured directly rather than taken from specifications.

Measurements were performed by members of the development team; no participants in the target age range contributed. Viewing distance and illuminance were not recorded.

C.2 Recorded accuracy runs

MediaPipe Iris with polynomial mapping, on the device above.

The two 16-point runs differ by approximately a factor of four in vertical error, which identifies run-to-run stability rather than error magnitude as the limiting factor.

Table C.1: Recorded accuracy runs

Configuration	Overall median	Vertical median	Note
9 points, unregularized	1.69 cm	1.69 cm	Run 1
9 points, unregularized	2.27 cm	2.06 cm	Run 2
16 points with ridge	2.08 cm	0.60 cm	Run 1
16 points with ridge	—	2.30 cm	Run 2, same setup minutes later
16 points, integrated build	1.68–2.42 cm	0.67–1.01 cm	Three runs

C.3 Retrained GAZEL

Trained from scratch on the TabletGaze corpus, evaluated on held-out subjects with a five-point per-user calibration.

Table C.2: Retrained GAZEL accuracy against training-set size

Subjects used for training	Calibrated error	Change
14	4.28 cm	—
28	3.89 cm	9 percent
37 (entire corpus)	3.87 cm	0.5 percent

A separate regularization pass, applied to address overfitting, made the calibrated result slightly worse. Both standard remedies for an underperforming model were therefore attempted without effect.

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